

# Low-Lying Zeros of Dirichlet $L$ -Functions

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## Abstract

In this paper, we investigate the relationship between the conductor  $q$  of a primitive Dirichlet  $L$ -function  $L(\chi, s)$  and the height of low-lying zeros. In particular, we use the explicit formula that relates the zeros of  $L(\chi, s)$  to a certain prime sum to carry out estimations with certain test functions. Assuming GRH, we observe that  $N(2, \chi) \geq 1$  for  $q \geq 262$  assuming GRH and use the techniques developed to give a more refined bound on gaps between consecutive zeros of Dirichlet  $L$  functions in terms of their conductor.

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# 1 Introduction

We investigate low-lying zeros of Dirichlet L functions with the aim of making explicit estimates about them. Throughout the text, let  $\chi$  denote a primitive character with conductor  $q > 1$  and  $\rho = \beta + i\gamma$  will denote zeros of Dirichlet-L functions  $L(s, \chi)$ . We also let

$$Z(\chi) = \{\rho \in C : 0 < \beta < 1, L(\rho, \chi) = 0\}$$

$$N(T, \chi) = |\{\rho \in Z(\chi) : |\gamma| \leq T\}|$$

We are mostly interested in the following question

**Question 1.1.** Let  $T > 0$ , what is an explicit  $q_0 \in \mathbb{N}$  such that for all primitive characters  $\chi$  with conductor  $q > q_0$ ,  $N(T, \chi) \geq 1$

To find the explicit value of  $q_0$ , we start with Backlund's argument to locate low-lying zeros of the Riemann-zeta function and extend it to various explicit formulas for general Dirichlet-L functions. In particular we work with the following explicit formula:

$$\phi(0) \log\left(\frac{q}{\pi}\right) = \sum_{\rho \in Z(\chi)} \Phi(\sigma - \rho) + \sum_{n \in \mathbb{N}} \Lambda(n) \left[ \frac{\chi(n) \phi(\log(n))}{n^\sigma} + \frac{\bar{\chi}(n) \phi(-\log(n))}{n^{1-\sigma}} \right]$$

$$- \frac{1}{2\pi i} \int_{1/2-i\infty}^{1/2+i\infty} \Re \frac{\Gamma'}{\Gamma}\left(\frac{s+\kappa}{2}\right) \Phi(\sigma-s) ds$$

Choosing a test function  $f$  such that its Laplace transform  $F(\sigma - \rho)$  is small for low-lying zeros, we estimate both sides of the equation and conclude that if there are no zeros below a certain height, the left-hand side stays small while the prime sum on the right-hand side is big, providing a contradiction.

## 1.1 Statement of Results

**Theorem 1.1.** For all  $q \geq q_0$  where  $q_0 = e^{79.866608}$ ,  $N(1, \chi) \geq 1$ .

Again, this theorem improves the trivial use of Theorem 1.1 of [1] for  $T = 1$  for obtaining the same result. That paper yields approximately  $q_0 = e^{108.45259}$ .

**Theorem 1.2.** Assuming GRH, then we have that for  $q \geq q_0$ ,  $N(T, \chi) \geq 1$

| $T$ | $q_0$               |
|-----|---------------------|
| 4   | 61.2526             |
| 3   | 160.1114            |
| 2   | 2500.3259           |
| 1   | 10,445,062,618.4334 |

Using numerical computation, we arrive at the following corollary

**Corollary 1.3.** Assuming GRH, then we have that for  $q \geq q_0$ ,  $N(T, \chi) \geq 1$  for the corresponding  $q_0$  and  $T$  in the table

| $T$ | $q_0$ |
|-----|-------|
| 4   | 6     |
| 3   | 26    |
| 2   | 242   |

Moreover,  $q_0$  is the smallest possible.

Theorem 1.2 parallels with [9, (6)], in which he covered explicitly  $T = 1$  and  $T = 5/7$ . We remark that although slight improvements can be made to our result, the result for  $T = 1$  in [9, p. 6] remains unbeaten.

Also, note that  $T = 4, T = 3$  is not of interest here because you can get those result using just classic formula for  $N(T, \chi)$ . The only thing our approach helped computed and is of interest is of  $T = 2$ . The result in [1] gives  $q_0 = 11606664.3031$ , for  $T = 2$ , which might be hard to compute up to, as the naive algorithm to get the smallest zeros is  $O(q^2)$ .

Originally the goal of this project is to compute  $N(1, \chi)$  also, but without improving theorem 1.2 or find a better algorithm than the naïve way of computing the the lowest zero of each L function  $L(s, \chi)$  for each  $\chi$  for each  $q$ , it is foreseeable that the computation will be expensive. The better value for  $T = 1$  [9] seems not good enough either.

Finally, we also have the following theorem.

**Theorem 1.4.** Assume GRH and let  $\dots \leq \gamma_{\chi, -2} \leq \gamma_{\chi, -1} \leq 0 \leq \gamma_{\chi, 0} \leq \gamma_{\chi, 1} \leq \dots$  be the zeros of the Dirichlet L function associated with conductor  $\chi$ . Then for a fixed  $n$ , fixed  $\epsilon > 0$ , for  $q$  sufficiently large, we have that

$$\gamma_{\chi, n+1} - \gamma_{\chi, n} \leq \pi(1 + \epsilon) \frac{1}{\log \log q - \log(4 + \frac{8}{\epsilon} + (1 + 2/\epsilon)O_n(1))}$$

In fact

$$\gamma_{\chi, n+1} - \gamma_{\chi, n} \leq (\pi + o(1)) \frac{1}{\log \log q} \quad q \rightarrow \infty$$

The bound  $\gamma_{\chi, n+1} - \gamma_{\chi, n} \ll \frac{1}{\log \log q}$  can be easily derived from the Riemann von Mongoldt formula and a GRH bound on  $S$  as mentioned in [10]. In fact, this theorem can be derived from [2, Example 8]. Moreover, it also follows from , I believe the coefficient with  $\pi(1 + \epsilon)$  in front is new, in the sense that it is not published anywhere explicitly.

## 2 Explicit Formulas for Dirichlet $L$ -Functions

We prove three different versions of the explicit formula in this section. In the next section, we will work with the last version and pick specific test functions.

**Proposition 2.1.** Let  $\varphi : (0, \infty) \rightarrow \mathbb{C}$  be a smooth function with compact support and

$$\hat{\varphi}(s) = \int_0^\infty \varphi(x) x^{s-1} dx$$

be its Mellin transform. Put  $\psi(x) = \frac{1}{x} \varphi(\frac{1}{x})$ . Then we have

$$\sum_n (\chi(n)\Lambda(n)\varphi(n) + \bar{\chi}(n)\Lambda(n)\psi(n)) = \varphi(1) \log q + r \int_0^\infty \varphi(x) dx +$$

$$\frac{1}{2\pi i} \int_{1/2-i\infty}^{1/2+i\infty} \left( \frac{\gamma'}{\gamma}(\chi, s) + \frac{\gamma'}{\gamma}(\bar{\chi}, 1-s) \right) \hat{\varphi}(s) ds - \sum_\rho \hat{\varphi}(\rho)$$

where  $r$  is the order of the pole of  $L(\chi, s)$  at  $s = 1$ , so  $r = 1$  if  $\chi = \chi_0$  is the principal character modulo  $q$  and  $r = 0$  if  $\chi \neq \chi_0$ .

*Proof.* Since  $\varphi$  is compactly supported  $\hat{\varphi}$  has rapid decay on each vertical strip  $(c-i\infty, c+i\infty)$  and it is analytic on all of  $\mathbb{C}$  and  $\int_{(c)} \hat{\varphi}(s) ds$  converges absolutely. Therefore, we can apply the inverse Mellin transform to write

$$\varphi(x) = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} \hat{\varphi}(s) x^{-s} ds$$

Then we see that

$$\begin{aligned} \sum_{n=1}^\infty \chi(n)\Lambda(n)\varphi(n) &= \frac{1}{2\pi i} \sum_n \chi(n)\Lambda(n) \int_{2-i\infty}^{2+i\infty} \hat{\varphi}(s) n^{-s} ds \\ &= \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} \left( \sum_n \chi(n)\Lambda(n) n^{-s} \right) \hat{\varphi}(s) ds \\ &= \frac{1}{2\pi i} \int_{(2)} -\frac{L'}{L}(\chi, s) \hat{\varphi}(s) ds \end{aligned}$$

Now we want to choose  $c > 0$  small enough so that  $L(\chi, s)$  does not have any zero in the strip  $-c < \Re s < 0$ . Choosing such a  $c$  is possible because the trivial zeros of  $L(\chi, s)$  are located at points  $-2m - \kappa$  for  $m \in \mathbb{N} - 0$  where  $\kappa = \begin{cases} 1 & \chi(-1) = -1 \\ 0 & \chi(-1) = 1 \end{cases}$ . Now, we apply the

Residue theorem to push the line of integration to the left, onto the strip  $(-c-i\infty, -c+i\infty)$  so that we can pick up residues from the zeros of  $L(\chi, s)$  inside the critical strip and from the pole at  $s = 1$ . The top and bottom edges of the rectangle vanish as we take them to  $\pm\infty$  because  $\left| -\frac{L'}{L}(\chi, s) \hat{\varphi}(s) \right|$  decays rapidly as  $\Im s \rightarrow \pm\infty$ . (Yes  $\hat{\varphi}(x)$  has rapid decay but why does it overpower the  $L$ -term?) After performing the shift of the integral, we obtain

$$\frac{1}{2\pi i} \int_{(2)} (\dots) = \frac{1}{2\pi i} \int_{(-c)} (\dots) + \sum (\text{residues})$$

Now, notice that if  $L$  has a zero of order  $m$  at  $s = \rho$ , then

$$L(\chi, s) = (s - \rho)^m \cdot g(s)$$

where  $g(s)$  is holomorphic and non-zero near  $\rho$ . Taking the logarithmic derivative,

$$-\frac{L'}{L}(\chi, s) = \frac{-m}{(s - \rho)} + G(s)$$

where  $G(s) = \frac{g'(s)}{g(s)}$  is holomorphic near  $\rho$ . So each zero  $\rho$  of  $L(\chi, s)$  contribute a residue of  $-m\hat{\varphi}(\rho)$ . Note that  $\hat{\varphi}(x)$  is analytic so it doesn't contribute a pole. Also,  $L(\chi, s)$  has a pole at  $s = 1$  of order  $r$  so the same calculation shows that we have a residue equal to  $r\hat{\varphi}(1) = r \int_0^\infty \varphi(x) dx$ . Hence, we have

$$\frac{1}{2\pi i} \int_{(2)} (\dots) = \frac{1}{2\pi i} \int_{(-c)} (\dots) + r \int_0^\infty \varphi(x) dx - \sum_{\rho} \varphi(\rho)$$

Now, we use the functional equation for  $L(\chi, s)$  which is given by

$$\Lambda(\chi, s) = \epsilon \cdot \Lambda(\bar{\chi}, 1 - s)$$

where

$$\begin{aligned} \Lambda(\chi, s) &= q^{s/2} \cdot \gamma(\chi, s) \cdot L(\chi, s) \\ &= \left(\frac{q}{\pi}\right)^{s/2} \Gamma\left(\frac{s + \kappa}{2}\right) \cdot L(\chi, s) \\ \epsilon &= \begin{cases} \frac{\tau(\chi)}{\sqrt{q}} & \chi(-1) = 1 \\ \frac{\tau(\chi)}{i\sqrt{q}} & \chi(-1) = -1 \end{cases} \end{aligned}$$

and  $\tau(\chi)$  is given by the Gaussian sum

$$\tau(\chi) = \sum_{m \in (\mathbb{Z}/q\mathbb{Z})} \chi(m) e^{2\pi i \frac{m^2}{q}}$$

Luckily,  $\epsilon$  does not depend on  $s$  so doesn't appear in the logarithmic derivative. Taking the logarithmic derivative of both sides,

$$-\frac{L'}{L}(\chi, s) = \log q + \frac{\gamma'}{\gamma}(\chi, s) + \frac{\gamma'}{\gamma}(\bar{\chi}, 1 - s) + \frac{L'}{L}(\bar{\chi}, 1 - s)$$

Integrating  $-\frac{L'}{L}(\chi, s)\hat{\varphi}(s)$  over the strip  $(-c)$ , the first term we get after plugging in the above formula is

$$\frac{1}{2\pi i} \int_{(-c)} \log q \cdot \hat{\varphi}(s) ds = \log q \cdot \varphi(1)$$

On the other hand, since  $\int_{(-c)} \hat{\varphi}(s) ds$  converges absolutely, we can rewrite

$$\begin{aligned}
\frac{1}{2\pi i} \int_{(-c)} \frac{L'}{L}(\bar{\chi}, 1-s) \hat{\varphi}(s) ds &= - \sum_{n=1}^{\infty} \bar{\chi}(n) \Lambda(n) \frac{1}{2\pi i} \int_{(-c)} \hat{\varphi}(s) n^{s-1} ds \\
&= - \sum_{n=1}^{\infty} \bar{\chi}(n) \Lambda(n) \frac{1}{2\pi i} \int_{(-c)} \hat{\psi}(1-s) n^{-(1-s)} ds \\
&= \sum_{n=1}^{\infty} \bar{\chi}(n) \Lambda(n) \frac{1}{2\pi i} \int_{c+1-i\infty}^{c+1+i\infty} \hat{\psi}(s) n^{-s} ds \\
&= - \sum_{n=1}^{\infty} \bar{\chi}(n) \Lambda(n) \frac{1}{2\pi i} \int_{c+1-i\infty}^{c+1+i\infty} \hat{\psi}(s) n^{-s} ds \\
&= - \sum_{n=1}^{\infty} \bar{\chi}(n) \Lambda(n) \psi(n)
\end{aligned}$$

Finally, the remaining integral is

$$\frac{1}{2\pi i} \int_{(-c)} \left( \frac{\gamma'}{\gamma}(\chi, s) + \frac{\gamma'}{\gamma}(\bar{\chi}, 1-s) \right) \hat{\varphi}(s) ds$$

Again, we can bring the line of integration to  $(\frac{1}{2})$  without picking up any residues because we have

$$\begin{aligned}
-\frac{\gamma'}{\gamma}(\bar{\chi}, 1-s) &= -\frac{d}{ds} \log \gamma(\bar{\chi}, 1-s) \\
&= \frac{d}{du} \log \gamma(\chi, u) \\
&= \frac{\gamma'}{\gamma}(\chi, u)
\end{aligned}$$

so the residues of  $\frac{\gamma'}{\gamma}(\chi, u)$  cancel out those coming from  $\frac{\gamma'}{\gamma}(\bar{\chi}, 1-s)$ .

After rearranging everything, we get

$$\begin{aligned}
\sum_n (\chi(n) \Lambda(n) \varphi(n) + \bar{\chi}(n) \Lambda(n) \psi(n)) &= \varphi(1) \log q + r \int_0^{\infty} \varphi(x) dx + \\
&\quad \frac{1}{2\pi i} \int_{1/2-i\infty}^{1/2+i\infty} \left( \frac{\gamma'}{\gamma}(\chi, s) + \frac{\gamma'}{\gamma}(\bar{\chi}, 1-s) \right) \hat{\varphi}(s) ds - \sum_{\rho} \hat{\varphi}(\rho)
\end{aligned}$$

as desired.  $\square$

We can improve the previous result by making a suitable change of variable to obtain Fourier transforms.

**Proposition 2.2.** Let  $h(x)$  be an *even* Schwartz function on  $\mathbb{R}$  and let  $\hat{h}(\xi)$  denote its Fourier transform. Then we have

$$\sum_{n=1}^{\infty} \Lambda(n)(\chi(n) + \bar{\chi}(n)) \cdot \frac{h(\log n)}{\sqrt{n}} = h(0) \cdot \log q + r \cdot h\left(\frac{i}{4\pi}\right) + \frac{1}{2\pi} \int_{-\infty}^{\infty} \left( \frac{\gamma'}{\gamma}(\chi, \frac{1}{2} + it) + \frac{\gamma'}{\gamma}(\bar{\chi}, \frac{1}{2} - it) \right) \cdot \hat{h}\left(\frac{t}{2\pi}\right) dt - \sum_{\rho} h\left(\frac{\gamma}{2\pi}\right)$$

where  $\rho = \frac{1}{2} + i\gamma$  runs over the zeros of  $L(\chi, s)$  on the critical line  $\Re s = \frac{1}{2}$ .

*Proof.* We make the following substitution to the setting in proposition 3.1

$$\varphi(x) = \frac{h(\log x)}{x^{1/2}} \quad x = e^y$$

With this, we have

$$\begin{aligned} \psi(x) &= \frac{1}{x} \varphi(1/x) \\ &= \frac{h(-\log x)}{x^{1/2}} \\ &= \frac{h(\log x)}{x^{1/2}} \quad (h \text{ is even}) \\ &= \varphi(x) \end{aligned}$$

On the other hand,

$$\begin{aligned} \hat{h}\left(\frac{i}{4\pi}\right) &= \int_{-\infty}^{\infty} h(x) e^{-2\pi i x \cdot \frac{i}{4\pi}} dx \\ &= \int_{-\infty}^{\infty} h(x) e^{x/2} dx \\ &= \int_0^{\infty} h(\log y) e^{-y/2} dy \\ &= \int_0^{\infty} \varphi(x) dx \end{aligned}$$

and  $\varphi(1) = h(\log 1) = h(0)$  and note that since  $h$  is even, so is its Fourier transform  $\hat{h}$ .

Hence,

$$\begin{aligned}
\hat{h}\left(\frac{t}{2\pi}\right) &= \hat{h}\left(\frac{-t}{2\pi}\right) \\
&= \int_{-\infty}^{\infty} h(x) e^{-2\pi i x \cdot \frac{-t}{2\pi}} dx \\
&= \int_{-\infty}^{\infty} h(x) e^{ixt} dx \\
&= \int_0^{\infty} h(\log y) \cdot y^{it} \frac{dy}{y} \\
&= \int_0^{\infty} \varphi(y) \cdot y^{it-1/2} dy \\
&= \hat{\varphi}\left(\frac{1}{2} + it\right)
\end{aligned}$$

And the same calculation as above shows that

$$\hat{\varphi}(\rho) = \varphi\left(\frac{1}{2} + i\gamma\right) = \hat{h}\left(\frac{\gamma}{2\pi}\right)$$

Plugging in these equivalent expressions to the main equation in proposition 3.1 yields

$$\begin{aligned}
\sum_{n=1}^{\infty} \Lambda(n)(\chi(n) + \overline{\chi}(n)) \cdot \frac{h(\log n)}{\sqrt{n}} &= h(0) \cdot \log q + r \cdot \hat{h}\left(\frac{i}{4\pi}\right) + \\
&\quad \frac{1}{2\pi} \int_{-\infty}^{\infty} \left( \frac{\gamma'}{\gamma}(\chi, \frac{1}{2} + it) + \frac{\gamma'}{\gamma}(\overline{\chi}, \frac{1}{2} - it) \right) \cdot \hat{h}\left(\frac{t}{2\pi}\right) dt - \sum_{\rho} h\left(\frac{\gamma}{2\pi}\right)
\end{aligned}$$

as desired. □

**Proposition 2.3.** Let  $f : \mathbb{R} \rightarrow \mathbb{C}$  be smooth and  $\sigma > 0.5$  be a real number. Denote by  $F$  the two-sided Laplace transform of  $f$ . That is,

$$F(s) = \int_{-\infty}^{\infty} f(t) e^{-st} dt$$

Then

$$\begin{aligned}
\sum_n \frac{\Lambda_{\chi}(n) f(\log n)}{n^{\sigma}} + \frac{\overline{\Lambda_{\chi}}(n) f(-\log n)}{n^{1-\sigma}} &= r \cdot F(\sigma - 1) - \sum_{\rho} F(\sigma - \rho) + f(0) \cdot \log q + \\
&\quad \frac{1}{2\pi i} \int_{(1/2)} \left[ \frac{\gamma'}{\gamma}(\chi, s) + \frac{\gamma'}{\gamma}(\overline{\chi}, 1 - s) \right] F(\sigma - s) ds \\
\sum_{\rho} F(\sigma - \rho) &= \sum_n \frac{\Lambda_{\chi}(n) f(\log n)}{n^{\sigma}} + \frac{\overline{\Lambda_{\chi}}(n) f(-\log n)}{n^{1-\sigma}} + r \cdot F(\sigma - 1) + f(0) \cdot \log q + \\
&\quad \frac{1}{2\pi i} \int_{(1/2)} \left[ \frac{\gamma'}{\gamma}(\chi, s) + \frac{\gamma'}{\gamma}(\overline{\chi}, 1 - s) \right] F(\sigma - s) ds
\end{aligned}$$

*Proof.* Start by using the inverse Laplace transform to write

$$\sum_n \Lambda_\chi(n) \frac{f(\log n)}{n^\sigma} = \frac{1}{2\pi i} \int_{(-\sigma)} \left( \sum_n \Lambda_\chi(n) n^u \right) F(u) du$$

Change variables to  $u \mapsto \sigma - s$  to rewrite

$$\frac{1}{2\pi i} \int_{(-\sigma)} \left( \sum_n \Lambda_\chi(n) n^u \right) F(u) du = \frac{1}{2\pi i} \int_{(2\sigma)} \left( \sum_n \Lambda_\chi(n) n^{-s} \right) F(\sigma - s) ds = \frac{1}{2\pi i} \int_{(2\sigma)} -\frac{L'}{L}(\chi, s) F(\sigma - s) ds$$

Moving the line of integration to  $(-c)$  where there are no zeroes of  $L(\chi, s)$  in the region  $-c < \Re s < 0$ , we pick up residues

$$-\sum_\rho F(\sigma - \rho) + rF(\sigma - 1)$$

and using the functional equation we can write

$$\frac{1}{2\pi i} \int_{(-c)} -\frac{L'}{L}(\chi, s) F(\sigma - s) ds = \frac{1}{2\pi i} \int_{-c} \left[ \log q + \frac{\gamma'}{\gamma}(\chi, s) + \frac{\gamma'}{\gamma}(\bar{\chi}, 1 - s) + \frac{L'}{L}(\bar{\chi}, 1 - s) \right] F(\sigma - s) ds$$

After a change of variable, the first term contributes  $f(0) \cdot \log q$ . Finally,

$$\begin{aligned} \frac{1}{2\pi i} \int_{(-c)} \frac{L'}{L}(\bar{\chi}, 1 - s) F(\sigma - s) ds &= -\sum_n \Lambda_{\bar{\chi}}(n) \frac{1}{2\pi i} \int_{(-c)} F(\sigma - s) n^{s-1} ds \\ &= -\sum_n \frac{\overline{\Lambda_\chi}(n) \cdot f(-\log n)}{n^{1-\sigma}} \end{aligned}$$

Finally, rearranging and pushing the line of integration for the  $\gamma$ -terms from  $(-c)$  to  $(1/2)$ . This change of contour does not pick-up any residues due to cancellations.

$$\begin{aligned} \sum_n \frac{\Lambda_\chi(n) f(\log n)}{n^\sigma} + \frac{\overline{\Lambda_\chi}(n) f(-\log n)}{n^{1-\sigma}} &= r \cdot F(\sigma - 1) - \sum_\rho F(\sigma - \rho) + f(0) \cdot \log q + \\ &\quad \frac{1}{2\pi i} \int_{(1/2)} \left[ \frac{\gamma'}{\gamma}(\chi, s) + \frac{\gamma'}{\gamma}(\bar{\chi}, 1 - s) \right] F(\sigma - s) ds \end{aligned}$$

□

Rewriting the above proposition a little bit and relaxing the constraint on  $f$ , we obtain the following

**Theorem 2.4.** Let  $\phi$  be a complex valued function and  $\sigma \in \mathbb{R}$  such that:

- (a)  $\phi$  is  $C^1$  on  $\mathbb{R} \setminus S$  where  $S$  is a finite set of points where for every  $a \in S$ , both  $\phi$  and  $\phi'$  have at worst jump discontinuities, and  $\phi(a) = \frac{\phi(a^+) + \phi(a^-)}{2}$ .
- (b) For some fixed  $b > 0$ , both the ROC of the Laplace transform of  $\phi$  and  $\phi'$  contains the vertical strip  $\sigma - (1 + b) < \Re(s) < \sigma + b$ .

Then if  $\Phi$  denotes the Laplace transform of  $\phi$  and  $\kappa = 0$  if  $\chi(-1) = 1$  and  $\kappa = 1$  if  $\chi(-1) = -1$ , we have

$$\begin{aligned} \phi(0) \log\left(\frac{q}{\pi}\right) &= \sum_{\rho \in Z(\chi)} \Phi(\sigma - \rho) + \sum_{n \in \mathbb{N}} \Lambda(n) \left[ \frac{\chi(n)\phi(\log(n))}{n^\sigma} + \frac{\bar{\chi}(n)\phi(-\log(n))}{n^{1-\sigma}} \right] \\ &\quad - \frac{1}{2\pi i} \int_{1/2-i\infty}^{1/2+i\infty} \Re \frac{\Gamma'}{\Gamma}\left(\frac{s+\kappa}{2}\right) \Phi(\sigma - s) ds \end{aligned} \quad (1)$$

Using the above, we obtain a version of Proposition 2.2

**Corollary 2.5.** Let  $\sigma \in \mathbb{R}$  and  $b > 0$  such that  $\Phi$  is analytic in  $\sigma - (1+b) < \Re(s) < \sigma + b$  and  $\Phi = O(\frac{1}{(1+|s|)^{1+\epsilon}})$  in the region for some  $\epsilon > 0$ . Then if  $\phi$  denotes the Laplace inversion, eq. (1) holds. Moreover, if we assume GRH, let  $\sigma = 1/2$  and let  $F(z) = \Phi(iz)$  (so this is a function defined on  $-1/2 + b < \Im(s) < 1/2 + b$ ), and so  $\phi(t) = \frac{1}{2\pi i} \int_{-i\infty}^{i\infty} e^{st} \Phi(s) ds = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ist} \Phi(is) ds = \check{F}(t)$ . Then we get that

$$\begin{aligned} \frac{\hat{F}(0)}{2\pi} \log\left(\frac{q}{\pi}\right) &= \sum_{\beta+i\gamma \in Z(\chi)} F(\gamma) + \frac{1}{2\pi} \sum_{n \in \mathbb{N}} \Lambda(n) \left[ \frac{\chi(n)\hat{F}(\log(n))}{\sqrt{n}} + \frac{\bar{\chi}(n)\hat{F}(-\log(n))}{\sqrt{n}} \right] \\ &\quad - \frac{1}{2\pi} \int_{-\infty}^{\infty} \Re \frac{\Gamma'}{\Gamma}\left(\frac{1/2+\kappa}{2} + \frac{it}{2}\right) F(t) dt \end{aligned} \quad (2)$$

*Proof.* Using these hypothesis, the proof of eq. (1) is exactly the same. Plugging in  $F(z) = \Phi(iz)$ , we get

$$\begin{aligned} \check{F}(0) \log\left(\frac{q}{\pi}\right) &= \sum_{\beta+i\gamma \in Z(\chi)} F(-\gamma) + \sum_{n \in \mathbb{N}} \Lambda(n) \left[ \frac{\chi(n)\check{F}(\log(n))}{\sqrt{n}} + \frac{\bar{\chi}(n)\check{F}(-\log(n))}{\sqrt{n}} \right] \\ &\quad - \frac{1}{2\pi} \int_{-\infty}^{\infty} \Re \frac{\Gamma'}{\Gamma}\left(\frac{1/2+\kappa}{2} + \frac{it}{2}\right) F(-t) dt \end{aligned}$$

Then as  $\check{F}(t) = \frac{1}{2\pi} \hat{F}(-t)$ , we get

$$\begin{aligned} \frac{\hat{F}(0)}{2\pi} \log\left(\frac{q}{\pi}\right) &= \sum_{\beta+i\gamma \in Z(\chi)} F(\gamma) + \frac{1}{2\pi} \sum_{n \in \mathbb{N}} \Lambda(n) \left[ \frac{\chi(n)\hat{F}(\log(n))}{\sqrt{n}} + \frac{\bar{\chi}(n)\hat{F}(-\log(n))}{\sqrt{n}} \right] \\ &\quad - \frac{1}{2\pi} \int_{-\infty}^{\infty} \Re \frac{\Gamma'}{\Gamma}\left(\frac{1/2+\kappa}{2} + \frac{it}{2}\right) F(t) dt \end{aligned}$$

□

Using the Gauss integral of the digamma function

$$\Re \frac{\Gamma'}{\Gamma}(z) = \int_0^\infty \frac{e^{-u}}{u} - \frac{e^{-\Re(z)u} \cos(\Im(z)u)}{1 - e^{-u}} du$$

We get

$$\begin{aligned}\Re \frac{\Gamma'}{\Gamma} \left( \frac{1+2\kappa}{4} + \frac{it}{2} \right) &= \int_0^\infty \frac{e^{-u}}{u} - \frac{e^{-(\frac{1+2\kappa}{4})u} \cos(\frac{t}{2}u)}{1-e^{-u}} du \\ &= \int_0^\infty \frac{e^{-u}}{u} - \frac{e^{-(\frac{1+2\kappa}{4})u} \cos(\frac{t}{2}u)}{1-e^{-u}} du\end{aligned}$$

So by Fubini's theorem, we have

$$\begin{aligned}-\frac{1}{2\pi} \int_{-\infty}^\infty \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2+\kappa}{2} + \frac{it}{2} \right) F(t) dt &= -\frac{1}{2\pi} \int_{-\infty}^\infty \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2+\kappa}{2} + \frac{it}{2} \right) F(t) dt \\ &= -\frac{1}{2\pi} \int_{-\infty}^\infty \int_0^\infty \frac{e^{-u}}{u} - \frac{e^{-(\frac{1/2+\kappa}{2})u} \cos(t/2u)}{1-e^{-u}} du F(t) dt \\ &= -\frac{1}{2\pi} \int_0^\infty \frac{e^{-u}}{u} \hat{F}(0) - \Re \hat{F} \left( \frac{u}{2} \right) \frac{e^{-(\frac{1/2+\kappa}{2})u}}{1-e^{-u}} du\end{aligned}$$

So we can rewrite eq. (2) as

$$\begin{aligned}\frac{\hat{F}(0)}{2\pi} \log \left( \frac{q}{\pi} \right) &= \sum_{\beta+i\gamma \in Z(\chi)} F(\gamma) + \frac{1}{2\pi} \sum_{n \in \mathbb{N}} \Lambda(n) \left[ \frac{\chi(n) \hat{F}(\log(n))}{\sqrt{n}} + \frac{\bar{\chi}(n) \hat{F}(-\log(n))}{\sqrt{n}} \right] \\ &\quad - \frac{1}{2\pi} \int_0^\infty \frac{e^{-u}}{u} \hat{F}(0) - \Re \hat{F} \left( \frac{u}{2} \right) \frac{e^{-(\frac{1/2+\kappa}{2})u}}{1-e^{-u}} du\end{aligned} \quad (3)$$

This now bares more resembles with the classic Weil explicit formula.

## 3 Explicit Computations

### 3.1 Classical Explicit Formula

Observe that theorem 2.4 is akin to the classical explicit formula by letting  $f$  be the Heaviside function. In this subsection, we toy with the classic explicit formula for answering question 1.1. In particular, letting  $\sigma > 1$

$$-\Re \frac{L'}{L}(\sigma, \chi) - \frac{1}{2} \psi \left( \frac{\sigma + \kappa}{2} \right) + \Re \sum_{\rho} \frac{1}{\sigma - \rho} = \frac{1}{2} \log \left( \frac{q}{\pi} \right)$$

Next, we start with estimating the term  $\sum_{\rho} \frac{1}{\sigma - \rho}$ . To do that, first need the following lemma

**Theorem 3.1.** Let  $T \geq 1$  and  $\chi$  a primitive character modulo  $q$ . Then

$$N(T, \chi) = \frac{T}{\pi} \log \left( \frac{qT}{2\pi e} \right) + O^*(C_1 \log(qT) + C_2) \quad (4)$$

with  $C_1 = 0.248$  and  $C_2 = 9.339$ . In addition  $C_1$  and  $C_2$  can be any constants in Table 1 of [7].

*Proof.* This is Theorem 1 of [7]. □

Using it, we can get a bound on the device  $\sum_{|\gamma| \geq T} \frac{1}{\gamma^2}$ .

**Lemma 3.2.** For  $T \geq 1$ , and  $\chi$  primitive non-principle Dirichlet character

$$\sum_{|\gamma| > T} \frac{1}{\gamma^2} = \frac{\log(\frac{qTe}{2\pi})}{\pi T} + O^*\left(\frac{2C_1 \log(qT) + 2C_2 + C_1/2}{T^2}\right)$$

*Proof.*

$$\begin{aligned} \sum_{|\gamma| > T} \frac{1}{\gamma^2} &= \int_T^\infty \frac{1}{t^2} dN(\cdot, \chi) \\ &= \lim_{U \rightarrow \infty} \frac{N(U, \chi)}{U^2} - \frac{N(T, \chi)}{T^2} + 2 \int_T^\infty \frac{N(t, \chi)}{t^3} dt \\ &= -\frac{N(T, \chi)}{T^2} + 2 \int_T^\infty \frac{N(t, \chi)}{t^3} dt \\ &= -\frac{\frac{T}{\pi} \log(\frac{qT}{2\pi e}) + O^*(C_1 \log(qT) + C_2)}{T^2} + 2 \int_T^\infty \frac{\frac{t}{\pi} \log(\frac{qt}{2\pi e}) + O^*(C_1 \log(qt) + C_2)}{t^3} dt \\ &= \frac{-\log(\frac{qT}{2\pi e})}{\pi T} + \frac{2 \log(\frac{qT}{2\pi})}{\pi T} + O^*\left(\frac{C_1 \log(qT) + C_2}{T^2}\right) + O^*\left(2 \int_T^\infty \frac{C_1 \log(qt) + C_2}{t^3} dt\right) \\ &= \frac{-\log(\frac{qT}{2\pi e})}{\pi T} + \frac{2 \log(\frac{qT}{2\pi})}{\pi T} + O^*\left(\frac{C_1 \log(qT) + C_2}{T^2}\right) + O^*\left(-\frac{2C_1 \ln(qt) + C_1}{2t^2} - \frac{C_2}{t^2} \Big|_T^\infty\right) \\ &= \frac{\log(\frac{qTe}{2\pi})}{\pi T} + O^*\left(\frac{2C_1 \log(qT) + 2C_2 + C_1/2}{T^2}\right) \end{aligned}$$

□

So in particular, we have that for  $\sigma > 1$ :

$$\begin{aligned} \Re \sum_{|\gamma| \geq T} \frac{1}{\sigma - \rho} &= \sum_{|\gamma| > T} \frac{\sigma - \beta}{(\sigma - \beta)^2 + \gamma^2} \\ &\leq \sum_{|\gamma| > T} \frac{\sigma}{\gamma^2} \\ &\leq \sigma \left( \frac{1}{\pi T} + \frac{2C_1}{T^2} \right) \log(q) + \frac{\log(\frac{e}{2\pi})}{\pi T} + \frac{2C_2 + C_1/2}{T^2} \end{aligned}$$

So if we have that  $\sigma$  and  $T$  satisfies  $\frac{1}{2} - \sigma(\frac{1}{\pi T} + \frac{2C_1}{T^2}) > 0$ , the classic explicit formula gives

$$-\Re \frac{L'}{L}(\sigma, \chi) - \frac{1}{2} \psi\left(\frac{\sigma + \kappa}{2}\right) + \frac{\log(\frac{e}{2\pi})}{\pi T} + \frac{2C_2 + C_1/2}{T^2} \geq \left(\frac{1}{2} - \sigma\left(\frac{1}{\pi T} + \frac{2C_1}{T^2}\right)\right) \log(q)$$

Then using the classic bound  $\log(x) - \frac{1}{x} \leq \psi(x) \leq \log(x) - \frac{1}{2x}$  and bounds on  $\frac{\zeta'}{\zeta}$  stated in [3, p. 1.4] writing  $\sigma = 1 + \epsilon$  gives

$$\begin{aligned} -\Re \frac{L'}{L}(\sigma, \chi) &= -\frac{\zeta'}{\zeta}(\sigma) \leq \frac{1}{\epsilon} \\ -\frac{1}{2} \Re \frac{\Gamma'}{\Gamma}\left(\frac{\sigma + \kappa}{2}\right) &< -\frac{1}{2} \log\left(\frac{\sigma + \kappa}{2}\right) + \frac{1}{\sigma + \kappa} \leq \frac{1}{2} \log\left(1 + \frac{\epsilon}{2}\right) + \frac{1}{1 + \epsilon} \end{aligned}$$

We conclude that if  $\sigma > 1$  and  $T \geq 1$  satisfies  $\frac{1}{2} - \sigma\left(\frac{1}{\pi T} + \frac{2C_1}{T^2}\right) > 0$ , we have that if

$$q > \exp\left(\frac{1}{1/2 - \sigma\left(\frac{1}{\pi T} + \frac{2C_1}{T^2}\right)} \cdot \left(\frac{1}{\epsilon} + \frac{1}{2} \log\left(1 + \frac{\epsilon}{2}\right) + \frac{1}{1 + \epsilon} + \frac{\log(e/2\pi)}{\pi T} + \frac{2C_2 + C_1/2}{T^2}\right)\right)$$

we get:

$$N(T, \chi) \geq 1$$

This leads to the following proposition.

**Proposition 3.3.** If  $q \geq \exp(64.45258)$ , then  $N(2, \chi) \geq 1$

*Proof.* Using  $C_1 = 0.248$  and  $C_2 = 9.339$ , and from a quick plot of the function

$$\epsilon \mapsto \exp\left(\frac{1}{1/2 - (1 + \epsilon)\left(\frac{1}{\pi T} + \frac{2C_1}{T^2}\right)} \cdot \left(\frac{1}{\epsilon} + \frac{1}{2} \log\left(1 + \frac{\epsilon}{2}\right) + \frac{\log(e/2\pi)}{\pi T} + \frac{1}{1 + \epsilon} + \frac{2C_2 + C_1/2}{T^2}\right)\right)$$

for  $T = 2$ , we see that at  $\epsilon = 0.23689$ ,  $1/2 - (1 + \epsilon)\left(\frac{1}{\pi T} + \frac{2C_1}{T^2}\right)$  is positive and the above is minimized at:

$$\exp(64.45258)$$

□

We remark that [1] has  $q \geq e^{14.0057}$ , then  $N(2, \chi) \geq 1$ , so this proposition is nothing impressive.

## 3.2 Non-positivity and Polynomials

Through our exploration above, we see that the sum over zeros term accounts for a good chunk of the error. However, if we let that assume that there are no-zeros of height less than  $T$ , and  $F$  satisfies that  $\Re F(s) \leq 0$  for  $\Re(s)$  in the strip in consideration and  $|\Im(s)| > T$ , we can introduce an upper bound on the main  $\log(q)$  term of Equation 1 by just dropping the entire sum over zero term.

This observation is also used in [9] for obtaining all the major theorems, though GRH was assumed so such functions are easier to find.

In this part, we see a simple example of the above mentioned method in action. Recall the classic explicit formula:

$$-\Re \frac{L'}{L}(\sigma, \chi) = \frac{1}{2} \psi\left(\frac{\sigma + \kappa}{2}\right) + \frac{1}{2} \log\left(\frac{q}{\pi}\right) - \Re \sum_{\rho} \frac{1}{\sigma - \rho}$$

Taking derivatives give

$$\begin{aligned} -\Re \frac{d}{ds} \frac{L'}{L}(\sigma, \chi) &= \frac{1}{4} \psi^{(1)}\left(\frac{\sigma + \kappa}{2}\right) + \Re \sum_{\rho} \frac{1}{(\sigma - \rho)^2} \\ -\Re \frac{d^2}{ds^2} \frac{L'}{L}(\sigma, \chi) &= \frac{1}{8} \psi^{(2)}\left(\frac{\sigma + \kappa}{2}\right) - 2\Re \sum_{\rho} \frac{1}{(\sigma - \rho)^3} \end{aligned}$$

Multiplying the second by  $-\sigma$  and the third by  $\frac{d}{2}$ , gives:

$$\begin{aligned} & -\Re \frac{L'}{L}(\sigma, \chi) - \frac{1}{2} \psi\left(\frac{\sigma + \kappa}{2}\right) + \sigma \Re \frac{d}{ds} \frac{L'}{L}(\sigma, \chi) + \sigma \frac{1}{2} \psi^{(1)}\left(\frac{\sigma + \kappa}{2}\right) - \frac{d}{2} \Re \frac{d^2}{ds^2} \frac{L'}{L}(\sigma, \chi) - \frac{d}{16} \psi^{(2)}\left(\frac{\sigma + \kappa}{2}\right) + \frac{\log(\pi)}{2} \\ &= \frac{1}{2} \log(q) - \Re \sum_{\rho} \left( \frac{1}{\sigma - \rho} + \sigma \frac{1}{(\sigma - \rho)^2} + d \frac{1}{(\sigma - \rho)^3} \right) \end{aligned}$$

Now, let  $F(s) = \frac{1}{s} + \sigma \frac{1}{s^2} + d \frac{1}{s^3} = \frac{f(0)}{s} + \frac{f'(0)}{s^2} + \frac{f''(0)}{s^3}$ . It is straight foward to check that

$$\Re(F(x + iy)) = \frac{f(0)x^5 + f'(0)x^4 + f''(0)x^3 + y^2(2x^3f(0) - 3xf''(0)) + y^4(f(0)x - f'(0))}{(x^2 + y^2)^3}$$

Then if we let  $d$  such that  $d > \frac{2}{3}\sigma^2$ , for  $0 < x < \sigma$ :

$$\begin{aligned} f(0)x - f'(0) &= x - \sigma < 0 \\ 2x^3f(0) - 3xf''(0) &= 2x^3 - 3xd < 0 \end{aligned}$$

Hence we get that  $\Re(F(x + iy)) \leq 0$  on  $0 < x < \sigma$  if we have:

$$|y| > \sqrt{\frac{f(0)x^5 + f'(0)x^4 + f''(0)x^3}{3xf''(0) - 2x^3f(0)}}$$

This condition is satisfied if:

$$|y| > \sigma \sqrt{\frac{f(0)\sigma^2 + f'(0)\sigma + f''(0)}{3f''(0) - 2\sigma^2f(0)}} = \sigma \sqrt{\frac{2\sigma^2 + d}{3d - 2\sigma^2}} = \sigma \sqrt{\frac{1 + \frac{2\sigma^2}{d}}{3 - \frac{2\sigma^2}{d}}}$$

Now, using the classic bound  $\frac{(m-1)!}{x^m} + \frac{m!}{2x^{m+1}} \leq (-1)^{m+1} \psi^{(m)}(x) \leq \frac{(m-1)!}{x^m} + \frac{m!}{x^{m+1}}$  for  $m \geq 1$  and  $x \geq 0$ , and [3, p. 3], [3, p. 12], we get

$$\begin{aligned}
-\Re \frac{L'}{L}(\sigma, \chi) &\leq \frac{1}{\sigma - 1} \\
|\sigma \Re \frac{d}{ds} \frac{L'}{L}(\sigma, \chi)| &\leq \sigma \frac{d}{ds} \frac{\zeta'}{\zeta}(\sigma) \leq \frac{\sigma}{(\sigma - 1)^2} \\
|\frac{d}{2} \Re \frac{d^2}{ds^2} \frac{L'}{L}(\sigma, \chi)| &\leq \frac{-d}{2} \frac{d^2}{ds^2} \frac{\zeta'}{\zeta}(\sigma) \leq \frac{d}{(\sigma - 1)^3} \\
-\frac{1}{2} \psi\left(\frac{\delta + \kappa}{2}\right) &< -\frac{1}{2} \log\left(\frac{\sigma + \kappa}{2}\right) + \frac{1}{\sigma + \kappa} < -\frac{1}{2} \log\left(\frac{\sigma}{2}\right) + \frac{1}{\sigma} \\
\frac{\sigma}{2} \psi^{(1)}\left(\frac{\sigma + \kappa}{2}\right) &< \frac{\sigma}{\sigma + \kappa} + \frac{2\sigma}{(\sigma + \kappa)^2} \leq 1 + \frac{2}{\sigma} \\
-\frac{d}{16} \psi^{(2)}\left(\frac{\sigma + \kappa}{2}\right) &< \frac{d}{4(\sigma + \kappa)^2} + \frac{d}{(\sigma + \kappa)^3} \leq \frac{d}{4\sigma^2} + \frac{d}{\sigma^3}
\end{aligned}$$

So all together, we have that:

$$\begin{aligned}
&\frac{1}{\sigma - 1} + \frac{\sigma}{(\sigma - 1)^2} + \frac{d}{(\sigma - 1)^3} - \frac{1}{2} \log\left(\frac{\sigma}{2}\right) + \frac{1}{\sigma} + 1 + \frac{2}{\sigma} + \frac{d}{4\sigma^2} + \frac{d}{\sigma^3} + \frac{1}{2} \log(\pi) \\
&\geq \frac{1}{2} \log(q)
\end{aligned}$$

Then we arrive at the following proposition

**Proposition 3.4.** For  $q \geq e^{362}$ ,  $N(1, \chi) \geq 1$ .

*Proof.* Using desmos with  $\sigma = 1.442823$  and  $d = 14$ . □

This proposition is again useless, see the remark under theorem 1.1.

### 3.2.1 Some Questions

Although the calculation above does not yield better numerics, it has benefit of being unconditional. This leads to the following questions.

**Question 3.1.** Does there exists a function  $f$  whose Laplace transform  $F$  is entire and  $\Re F(x + iy) \leq 0$  for all  $x \in \mathbb{R}$  and  $|y| \geq L$  for some fixed  $L$ .

If so, we can obtain the results in [9] using Equation 1 but unconditionally. For obtain numerics of the form of finding explicit  $q_0$  where  $q \geq q_0$  implies  $N(T, \chi) \geq 1$ , it is enough to answer the following.

**Question 3.2.** Does there exists a function  $f_T$  whose Laplace transform  $F_T$  has a region of absolute convergence containing the strip  $\sigma - 1/2 < \Re(s) < \sigma + 1/2$  and  $\Re F(x + iy) \leq 0$  for  $-1/2 + \sigma < x < 1/2 + \sigma$  and  $|y| \geq T$ .

Much time is devoted into finding such functions, though no significant breakthrough is achieved.

### 3.3 Gaussian Distribution

This section outlines the proof of theorem 1.1. For each  $R > 0$  and  $\mu \in \mathbb{R}$ , consider the Gaussian distribution with mean  $\mu$  and standard deviation  $\sqrt{2}R$

$$f(t) = f_{R,\mu}(t) = \frac{1}{2\sqrt{\pi}R} e^{-\frac{(t-\mu)^2}{4R^2}}$$

It is easy to compute the Laplace transform  $\mathcal{L}(f)$  of  $f$

**Proposition 3.5.** For all  $s \in \mathbb{C}$   $\mathcal{L}(f)(s) = e^{R^2 s^2 - \mu s}$ .

*Proof.* For  $s \in \mathbb{R}$ , we get

$$\begin{aligned} \mathcal{L}(f)(s) &= \int_{\mathbb{R}} \frac{1}{2\sqrt{\pi}R} e^{-\frac{(t-\mu)^2}{4R^2}} e^{-st} dt \\ &= \int_{\mathbb{R}} \frac{1}{2\sqrt{\pi}R} e^{-\frac{1}{4R^2}[(t-\mu)^2 + 4R^2 st]} dt \\ &= \int_{\mathbb{R}} \frac{1}{2\sqrt{\pi}R} e^{-\frac{1}{4R^2}[t^2 - 2(\mu - 2R^2 s)t + \mu^2]} dt \\ &= \int_{\mathbb{R}} \frac{1}{2\sqrt{\pi}R} e^{-\frac{1}{4R^2}[(t - (\mu - 2R^2 s))^2 - (\mu - 2R^2 s)^2 + \mu^2]} dt \\ &= \int_{\mathbb{R}} \frac{1}{2\sqrt{\pi}R} e^{-\frac{1}{4R^2}[(t - (\mu - 2R^2 s))^2]} e^{R^2 s^2 - \mu s} dt \\ &= e^{R^2 s^2 - \mu s} \int_{\mathbb{R}} \frac{1}{2\sqrt{\pi}R} e^{-\frac{1}{4R^2}[(t - (\mu - 2R^2 s))^2]} dt \\ &= e^{R^2 s^2 - \mu s} \end{aligned}$$

In the last line, we use the fact that the integrand is a Gaussian distribution with mean  $\mu - 2R^2 s$  and standard deviation  $\sqrt{2}R$ . Now, as  $f$  is Schwartz,  $\mathcal{L}(f)$  is entire. Then as  $e^{R^2 s^2 - \mu s}$  is also entire and they agree on  $\mathbb{R}$ , they also agree on  $\mathbb{C}$ .  $\square$

The next sequence of lemmas gives bounds on term of eq. (1).

#### 3.3.1 Studying the Prime Sum

**Proposition 3.6.** Let  $c = 1.03883$ , then

- If  $e^{\mu - 2\sigma R^2} \geq 2$ ,

$$\left| \sum_{n \geq 1} \frac{\Lambda(n) \chi(n) f(\log(n))}{n^\sigma} \right| \leq c \mathcal{L}(f)(\sigma - 1) \left( 1 + \frac{1}{2\sqrt{\pi}R} e^{-R^2} \right)$$

else if  $e^{\mu - 2\sigma R^2} < 2$

$$\left| \sum_{n \geq 1} \frac{\Lambda(n) \chi(n) f(\log(n))}{n^\sigma} \right| \leq c \frac{2^{1-\sigma}}{2\sqrt{\pi}R} e^{-\frac{(\log(2) + \mu)^2}{4R^2}} + c \mathcal{L}(f)(\sigma - 1)$$

- Similarly if  $e^{-\mu-2(1-\sigma)R^2} \geq 2$ :

$$\left| \sum_{n=2}^{\infty} \frac{\Lambda(n)\chi(n)f(-\log(n))}{n^{1-\sigma}} \right| \leq c\mathcal{L}(f)(-\sigma)(1 + \frac{1}{2\sqrt{\pi}R}e^{-R^2})$$

- If  $e^{-\mu-2(1-\sigma)R^2} < 2$ :

$$\left| \sum_{n=2}^{\infty} \frac{\Lambda(n)\chi(n)f(-\log(n))}{n^{1-\sigma}} \right| \leq c \frac{2^\sigma}{2\sqrt{\pi}R} e^{-\frac{(\log(2)+\mu)^2}{4R^2}} + c\mathcal{L}(f)(-\sigma)$$

*Proof.* Let  $w(x) = x^{-\sigma}f(\log(x))$ , observe that

$$\begin{aligned} w'(x) &= -\sigma x^{-\sigma-1}f(\log(x)) + x^{-\sigma-1}f'(\log(x)) \\ &= -\sigma x^{-\sigma-1}f(\log(x)) + x^{-\sigma-1}\left(-\frac{\log(x) - \mu}{2R^2}\right)f(\log(x)) \\ &= \frac{w(x)}{x}\left(-\sigma - \frac{\log(x) - \mu}{2R^2}\right) \\ &= -\frac{w(x)}{x}\left(\sigma + \frac{\log(x) - \mu}{2R^2}\right) \end{aligned}$$

So  $w'(x)$  is zero at  $e^{\mu-2\sigma R^2}$ , is positive before this point and negative after that point. Hence we get a global maximum there. So then using Theorem 12 (line 3.35) of [5], we get for  $c = 1.03883$ , for all  $x > 0$

$$\psi(x) < cx$$

Now, let  $x_0 = e^{\mu-2\sigma R^2}$ , then (should include more detail on the fact  $w(x) \rightarrow 0$  as  $x \rightarrow \infty$ )

- If  $e^{\mu-2\sigma R^2} \geq 2$

$$\begin{aligned}
\left| \sum_{n=2}^{\infty} \frac{\Lambda(n)\chi(n)f(\log(n))}{n^{1-\sigma}} \right| &\leq \sum_{n=2}^{\infty} \Lambda(n)w(n) \\
&= \int_{2^-}^{\infty} w(n)d\psi(n) \\
&= - \int_2^{\infty} w'(x)\psi(x)dx \\
&= - \int_2^{x_0} w'(x)\psi(x) - \int_{x_0}^{\infty} w'(x)\psi(x) \\
&\leq - \int_{x_0}^{\infty} w'(x)\psi(x) \\
&\leq -c \int_{x_0}^{\infty} xw'(x)dx \\
&= c[-xw(x)]_{x_0}^{\infty} + c \int_{x_0}^{\infty} w(x)dx \\
&= cx_0w(x_0) + c \int_{x_0}^{\infty} w(x)dx \\
&= ce^{(1-\sigma)(\mu-2\sigma R^2)} \frac{1}{2\sqrt{\pi}R} e^{-\sigma^2 R^2} + c \int_{\log(x_0)}^{\infty} e^{(1-\sigma)t} f(t)dt \\
&< c \frac{1}{2\sqrt{\pi}R} e^{-R^2} \mathcal{L}(f)(\sigma-1) + c\mathcal{L}(f)(\sigma-1) \\
&= c\mathcal{L}(f)(\sigma-1)(1 + \frac{1}{2\sqrt{\pi}R} e^{-R^2})
\end{aligned}$$

- If  $x_0 < 2$ , then looking at the computation above, we get:

$$\left| \sum_{n=2}^{\infty} \frac{\Lambda(n)\chi(n)f(\log(n))}{n^{1-\sigma}} \right| \leq c \frac{2^{1-\sigma}}{2\sqrt{\pi}R} e^{-\frac{(\log(2)+\mu)^2}{4R^2}} + c\mathcal{L}(f)(\sigma-1)$$

Observe that for  $\sum_{n=2}^{\infty} \frac{\Lambda(n)\overline{\chi(n)}f(-\log(n))}{n^{1-\sigma}}$  it is the same as above with  $\sigma = 1-\sigma$  and  $\mu = -\mu$ , thus we are done.  $\square$

### 3.3.2 Studying the Integral Term

**Lemma 3.7.** For all  $\sigma > 0$  the map  $t \mapsto \Re\psi(\sigma + it)$  is even and non-decreasing on  $[0, \infty)$ .

*Proof.* Let  $h(t) = \Re\psi(\sigma + it)$ . Since  $\psi$  is meromorphic and the Schwartz reflection principle (I believe)

$$\overline{\psi(\sigma + it)} = \psi(\sigma - it)$$

Therefore

$$h(t) = \Re\psi(\sigma + it) = \Re\overline{\psi(\sigma + it)} = \Re\psi(\sigma - it) = h(-t)$$

Next, note that

$$h'(t) = \Re i\psi^{(1)}(\sigma + it) = -\Im \psi^{(1)}(\sigma + it)$$

Then since

$$\psi^{(1)}(z) = \sum_{k=0}^{\infty} \frac{1}{(z+k)^2}$$

We get

$$h'(t) = -\Im \sum_{k=0}^{\infty} \frac{(\sigma - it)^2}{|\sigma + it + k|^4} = -\Im \sum_{k=0}^{\infty} \frac{\sigma^2 - t^2 - 2\sigma ti}{|\sigma + it + k|^4} = 2\sigma t \sum_{k=0}^{\infty} \frac{1}{|\sigma + it + k|^4} \geq 0$$

So we are done.  $\square$

**Proposition 3.8.**

$$\left| -\frac{1}{2\pi i} \int_{1/2-i\infty}^{1/2+i\infty} \Re \frac{\Gamma'}{\Gamma} \left( \frac{s+\kappa}{2} \right) \mathcal{L}f(\sigma-s) ds \right| \leq \frac{e^{R^2(\sigma-1/2)^2 - \mu(\sigma-1/2)}}{2\sqrt{\pi}R} \left( \frac{1}{2} \log((1/2+\kappa)^2 + \frac{1}{2R^2}) + \gamma - 2\psi\left(\frac{1+2\kappa}{4}\right) \right)$$

*Proof.* First, observe that

$$\begin{aligned} & \left| -\frac{1}{2\pi i} \int_{1/2-i\infty}^{1/2+i\infty} \Re \frac{\Gamma'}{\Gamma} \left( \frac{s+\kappa}{2} \right) \mathcal{L}f(\sigma-s) ds \right| \\ & \leq \frac{1}{2\pi} \int_{1/2-i\infty}^{1/2+i\infty} \left| \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2+\kappa+it}{2} \right) \mathcal{L}f(\sigma-1/2-it) \right| dt \\ & = \frac{1}{2\pi} \int_{1/2-i\infty}^{1/2+i\infty} \left| \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2+\kappa+it}{2} \right) e^{R^2[(\sigma-1/2)^2 - 2(\sigma-1/2)iu - t^2] - \mu(\sigma-1/2) + \mu(it)} \right| dt \\ & = \frac{e^{R^2(\sigma-1/2)^2 - \mu(\sigma-1/2)}}{2\pi} \int_{-\infty}^{\infty} \left| \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2+\kappa+it}{2} \right) \right| e^{-R^2 t^2} dt \end{aligned}$$

Let us first bound the  $\frac{\Gamma'}{\Gamma}$  term. By lemma 2.7 of [6], we have that  $\Re(z) \geq 1/2$  implies  $\Re \frac{\Gamma'}{\Gamma}(z/2) \leq \log(|z|) - \gamma$  and hence by lemma 3.7

$$\frac{\Gamma'}{\Gamma} \left( \frac{1+2\kappa}{4} \right) \leq \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2+\kappa+it}{2} \right) \leq \log\left(\left|\frac{1}{2}+\kappa+it\right|\right) - \gamma = \frac{1}{2} \log\left(\left(\frac{1}{2}+\kappa\right)^2 + t^2\right) - \gamma$$

$\square$

So

$$0 \leq \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2+\kappa+it}{2} \right) - \psi\left(\frac{1+2\kappa}{4}\right) \leq \frac{1}{2} \log((1/2+\kappa)^2 + t^2) - \gamma - \psi\left(\frac{1+2\kappa}{4}\right)$$

Therefore

$$\begin{aligned}
|\Re \frac{\Gamma'}{\Gamma}(\frac{1/2 + \kappa + it}{2})| &= |\Re \frac{\Gamma'}{\Gamma}(\frac{1/2 + \kappa + it}{2}) - \psi(\frac{1/2 + \kappa}{4}) + \psi(\frac{1/2 + \kappa}{4})| \\
&\leq \Re \frac{\Gamma'}{\Gamma}(\frac{1/2 + \kappa + it}{2}) - \psi(\frac{1/2 + \kappa}{4}) - \psi(\frac{1/2 + \kappa}{4}) \\
&\leq \frac{1}{2} \log((1/2 + \kappa)^2 + t^2) - \gamma - 2\psi(\frac{1 + 2\kappa}{4})
\end{aligned}$$

Now, let  $E[X] = \int_{\mathbb{R}} X(x) \frac{R}{\sqrt{\pi}} e^{-R^2 t^2} d\mu(x)$ . So we get

$$\begin{aligned}
&\left| -\frac{1}{2\pi i} \int_{1/2-i\infty}^{1/2+i\infty} \Re \frac{\Gamma'}{\Gamma}(\frac{s + \kappa}{2}) \mathcal{L}f(\sigma - s) ds \right| \\
&\leq \frac{e^{R^2(\sigma-1/2)^2 - \mu(\sigma-1/2)}}{2\sqrt{\pi}R} \int_{-\infty}^{\infty} \left[ \frac{1}{2} \log((1/2 + \kappa)^2 + t^2) - \gamma - 2\psi(\frac{1 + 2\kappa}{4}) \right] \frac{R}{\sqrt{\pi}} e^{-R^2 t^2} dt \\
&= \frac{e^{R^2(\sigma-1/2)^2 - \mu(\sigma-1/2)}}{2\sqrt{\pi}R} E \left[ \frac{1}{2} \log((1/2 + \kappa)^2 + t^2) - \gamma - 2\psi(\frac{1 + 2\kappa}{4}) \right] \\
&= \frac{e^{R^2(\sigma-1/2)^2 - \mu(\sigma-1/2)}}{2\sqrt{\pi}R} \frac{1}{2} E [\log((1/2 + \kappa)^2 + t^2)] - \gamma - 2\psi(\frac{1 + 2\kappa}{4}) \\
&\leq \frac{e^{R^2(\sigma-1/2)^2 - \mu(\sigma-1/2)}}{2\sqrt{\pi}R} \left( \frac{1}{2} E [\log((1/2 + \kappa)^2 + t^2)] - \gamma - 2\psi(\frac{1 + 2\kappa}{4}) \right) \\
&\leq \frac{e^{R^2(\sigma-1/2)^2 - \mu(\sigma-1/2)}}{2\sqrt{\pi}R} \left( \frac{1}{2} \log((1/2 + \kappa)^2 + E[t^2]) + \gamma - 2\psi(\frac{1 + 2\kappa}{4}) \right) \\
&= \frac{e^{R^2(\sigma-1/2)^2 - \mu(\sigma-1/2)}}{2\sqrt{\pi}R} \left( \frac{1}{2} \log((1/2 + \kappa)^2 + \frac{1}{2R^2}) + \gamma - 2\psi(\frac{1 + 2\kappa}{4}) \right)
\end{aligned}$$

### 3.3.3 Studying Sum over Zeros

**Lemma 3.9.** Let  $T \geq 1$ ,

$$\begin{aligned}
&\sum_{\rho \in Z(\chi), |\Im \rho| > T} e^{-R^2 \gamma^2} \\
&= \frac{1}{\pi R} \log\left(\frac{q}{2\pi R e}\right) \Gamma\left(\frac{3}{2}, R^2 T^2\right) + \frac{1}{2\pi R} \frac{\partial}{\partial s} \Gamma\left(\frac{3}{2}, R^2 T^2\right) - e^{-R^2 T^2} \frac{T}{\pi} \log\left(\frac{qT}{2\pi e}\right) \\
&\quad - e^{-R^2 T^2} O^*(C_1 \log(qT) + C_2) + O^*\left[(C_1 \log(qT) + C_2) e^{-R^2 T^2} + \frac{C_1}{2} E_1(R^2 T^2)\right] \\
&\leq \frac{1}{\pi R} \log\left(\frac{q}{2\pi R e}\right) \Gamma\left(\frac{3}{2}, R^2 T^2\right) + \frac{1}{2\pi R} \frac{\partial}{\partial s} \Gamma\left(\frac{3}{2}, R^2 T^2\right) \\
&\quad + \left[(C_1 \log(qT) + C_2) e^{-R^2 T^2} + \frac{C_1}{2} E_1(R^2 T^2)\right]
\end{aligned}$$

*Proof.* Let us first do  $T \geq 1$ , note that

$$\begin{aligned}
& \sum_{\rho \in Z(\chi), |\Im \rho| > T} e^{-R^2 \gamma^2} \\
&= \int_T^\infty e^{-R^2 t^2} dN(\chi) \\
&= -e^{-R^2 T^2} N(T, \chi) + \int_T^\infty 2R^2 t e^{-R^2 t^2} N(t, \chi) dt \\
&= -e^{-R^2 T^2} N(T, \chi) + \int_T^\infty 2R^2 t e^{-R^2 t^2} \left[ \frac{t}{\pi} \log\left(\frac{qt}{2\pi e}\right) + O^*(C_1 \log(qt) + C_2) \right] dt \\
&= -e^{-R^2 T^2} N(T, \chi) + \int_T^\infty 2R^2 t e^{-R^2 t^2} \frac{t}{\pi} \log\left(\frac{qt}{2\pi e}\right) dt + \int_T^\infty 2R^2 t e^{-R^2 t^2} O^*(C_1 \log(qt) + C_2) dt
\end{aligned}$$

Now, we will drop the first term there because it is negative (it produced better numerics at the end) Then note if  $E_1(x) = \int_x^\infty \frac{e^{-t}}{t} dt$  is the exponential integral, note for the error term

$$\begin{aligned}
& \int_T^\infty 2R^2 t e^{-R^2 t^2} O^*(C_1 \log(qt) + C_2) dt \\
&= O^* \int_T^\infty 2R^2 t e^{-R^2 t^2} (C_1 \log(qt) + C_2) dt \\
&= O^* \int_{R^2 T^2}^\infty e^{-u} \left( C_1 \log\left(\frac{q}{R}\right) + \frac{C_1}{2} \log(u) + C_2 \right) du \quad (u = R^2 t^2, du = 2R^2 t dt) \\
&= O^* \left[ \left( C_1 \log\left(\frac{q}{R}\right) + C_2 \right) e^{-R^2 T^2} + \frac{C_1}{2} (e^{-R^2 T^2} \log(R^2 T^2) + E_1(R^2 T^2)) \right] \\
&= O^* \left[ (C_1 \log(qT) + C_2) e^{-R^2 T^2} + \frac{C_1}{2} E_1(R^2 T^2) \right]
\end{aligned}$$

Finally for the main term

$$\begin{aligned}
\int_T^\infty 2R^2 t e^{-R^2 t^2} \left[ \frac{t}{\pi} \log\left(\frac{qt}{2\pi e}\right) \right] dt &= \frac{1}{\pi R} \int_{R^2 T^2}^\infty \sqrt{u} e^{-u} \left[ \log\left(\frac{q}{2\pi R e}\right) + \frac{1}{2} \log(u) \right] du \quad (u = R^2 t^2, du = 2R^2 t dt) \\
&= \frac{1}{\pi R} \log\left(\frac{q}{2\pi R e}\right) \Gamma\left(\frac{3}{2}, R^2 T^2\right) + \frac{1}{2\pi R} \frac{\partial}{\partial s} \Gamma\left(\frac{3}{2}, R^2 T^2\right)
\end{aligned}$$

We are done, as claimed.  $\square$

**Lemma 3.10.** If  $T \geq 1$ ,  $\sigma = \frac{1}{2}$  and  $\mu = 0$

$$\left| \sum_{\rho \in Z(\chi), |\Im \rho| > T} \mathcal{L}(\sigma - \rho) \right| \leq e^{\frac{R^2}{4}} \sum_{\rho \in Z(\chi), |\Im \rho| > T} e^{-R^2 \gamma^2}$$

*Proof.* The result follows from the computation

$$\begin{aligned}
\left| \sum_{\rho \in Z(\chi), |\Im \rho| > T} \mathcal{L}(\sigma - \rho) \right| &\leq \sum_{|\Im \rho| > T} |\mathcal{L}(\sigma - \rho)| \\
&\leq \sum_{|\Im \rho| > T} e^{(R^2[(\sigma - \beta)^2 - \gamma^2] - \mu(\sigma - \beta))} \\
&\leq e^{\frac{R^2}{4}} \sum_{|\Im \rho| > T} e^{-R^2 \gamma^2}
\end{aligned}$$

□

Finally, we get the main theorem

### 3.3.4 Proof of Theorem 1.1

*proof of Theorem 1.1.* Using the lemmas, we get for  $T \geq 1$ , assuming there are no zeros of height less than  $T$ , we get

$$\begin{aligned}
&\frac{1}{2\sqrt{\pi}R} \log(q) \\
&\leq \frac{1}{2\sqrt{\pi}R} \left( \frac{1}{2} \log((1/2 + \kappa)^2 + \frac{1}{2R^2}) + \gamma - 2\psi\left(\frac{1 + 2\kappa}{4}\right) \right) \\
&+ 2c \left[ \frac{1}{\sqrt{2}\sqrt{\pi}R} e^{-\frac{(\log(2))^2}{4R^2}} + \mathcal{L}(f)(1/2) \right] \\
&+ e^{\frac{R^2}{4}} \left( \frac{\Gamma(\frac{3}{2}, R^2 T^2)}{\pi R} + C_1 e^{-R^2 T^2} \right) \log(q) \\
&+ e^{\frac{R^2}{4}} \left[ \frac{1}{\pi R} \log\left(\frac{1}{2\pi R e}\right) \Gamma\left(\frac{3}{2}, R^2 T^2\right) + \frac{1}{2} \frac{\partial}{\partial s} \Gamma\left(\frac{3}{2}, R^2 T^2\right) + C_2 e^{-R^2 T^2} + C_1 \log(T) e^{-R^2 T^2} + \frac{C_1}{2} E_1(R^2 T^2) \right] \\
&+ \frac{1}{2\sqrt{\pi}R} \log(\pi)
\end{aligned}$$

Now, using Gemini 3.1 pro to generate some FLINT script to calculate the constants, we get the result for  $T = 1$  and  $R = 2.04$ .

□

## 3.4 Beurling Selburg Minorant

Let  $B(z)$  be Beurling's extremal majorant function of  $\text{sgn}(x)$ . It is defined as

$$B(z) = \left( \frac{\sin(\pi z)}{\pi} \right)^2 \left( \sum_{n=0}^{\infty} \frac{1}{(z - n)^2} - \sum_{m=-\infty}^{-1} \frac{1}{(z - m)^2} + \frac{2}{z} \right)$$

It satisfies that  $\text{sgn}(x) \leq B(x)$  for all  $x$  and

$$\int_{-\infty}^{\infty} B(x) - \text{sgn}(x) dx = 1$$

Letting  $b(x) = -B(-x)$ , we get that  $b(x) \leq \text{sgn}(x)$  for all  $x$  and

$$\int_{-\infty}^{\infty} \text{sgn}(x) - b(x) dx = 1$$

See [8] for more details. Now, fix  $\delta > 0$ ,  $\alpha < \beta$ , and let

$$m_{\delta, [\alpha, \beta]}(x) = \frac{1}{2}(b(\delta(\beta - z)) + b(\delta(z - \alpha)))$$

This is the Beurling-Selburg minorant of the indicator function  $\chi_{[\alpha, \beta]}$  with bandwidth  $2\pi\delta$ . It follows that  $m_{\delta, [a, b]}(x) \leq \chi_{[a, b]}(x)$  for all  $x \in \mathbb{R}$  and in fact

**Proposition 3.11.**

$$\widehat{m}_{\delta, [\alpha, \beta]}(0) = \beta - \alpha - \frac{1}{\delta}$$

*Proof.* Indeed

$$\begin{aligned} \widehat{m}_{\delta, [\alpha, \beta]}(0) &= \int_{-\infty}^{\infty} m_{\delta, [\alpha, \beta]}(x) dx \\ &= - \int_{-\infty}^{\infty} -m_{\delta, [\alpha, \beta]}(x) dx \\ &= - \int_{-\infty}^{\infty} \chi_{[-\alpha, \beta]}(x) - m_{\delta, [\alpha, \beta]}(x) dx + \int_{-\infty}^{\infty} \chi_{[\alpha, \beta]}(x) dx \\ &= -\frac{1}{2} \int_{-\infty}^{\infty} \text{sgn}(\beta - x) - b(\delta(\beta - x)) + \text{sgn}(x - \alpha) - b(\delta(x - \alpha)) dx + \int_{-\infty}^{\infty} \chi_{[\alpha, \beta]}(x) dx \\ &= -\frac{1}{\delta} + \beta - \alpha \end{aligned}$$

□

### 3.5 Proof of Theorem 1.2

It is not hard to see that  $B(z) = O(\frac{1}{|z|^2})$  as  $|z| \rightarrow \infty$ , so  $m_{\delta, [\alpha, \beta]}(z) = O(\frac{1}{|z|^2})$  as  $|z| \rightarrow \infty$  as well. Therefore, we can use eq. (2) and we get that

$$\begin{aligned} &\frac{\widehat{m}_{\delta, [-L, L]}(0)}{2\pi} \log\left(\frac{q}{\pi}\right) \\ &= \sum_{\beta + i\gamma} m_{\delta, [-L, L]}(\gamma) + \frac{1}{2\pi} \sum_{n \in \mathbb{N}} \Lambda(n) \left[ \frac{\chi(n) \widehat{m}_{\delta, [-L, L]}(\log(n))}{\sqrt{n}} + \frac{\overline{\chi}(n) \widehat{m}_{\delta, [-L, L]}(-\log(n))}{\sqrt{n}} \right] \\ &\quad - \frac{1}{2\pi} \int_{-\infty}^{\infty} \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa}{2} + \frac{it}{2} \right) m_{\delta, [-L, L]}(t) dt \end{aligned}$$

Since  $m_{\delta, [-L, L]}$  is even, we will bound the prime sum from above by

$$\frac{1}{2\pi} \sum_{n < e^{2\pi\delta}} \frac{\Lambda(n) |\widehat{m}_{\delta, [-L, L]}(\log(n))|}{\sqrt{n}}$$

In our application,  $\delta$  is not too big, so we can compute the above exactly using a computer. All we need is a formula for  $\widehat{m}_{\delta,[-L,L]}$ , which is the content of the following proposition.

**Proposition 3.12.** Let  $\delta > 0$  and  $L > 0$ , then

$$\widehat{m}_{\delta,[-L,L]} = \begin{cases} \frac{1}{\delta} \left( \left(1 - \frac{|x|}{2\pi\delta}\right) (\sin(Lx) \cot(\frac{x}{2\delta}) - \cos(Lx) + \frac{\sin(Lx)\operatorname{sgn}(x)}{\pi}) \right) & \text{if } |x| \leq 2\pi\delta \\ 0 & \text{otherwise} \end{cases}$$

*Proof.* Let

$$H(z) = \left( \frac{\sin \pi z}{\pi} \right)^2 \left( \sum_{m=-\infty}^{\infty} \operatorname{sgn}(m) \frac{1}{(z-m)^2} + \frac{2}{z} \right)$$

$$K(z) = \left( \frac{\sin \pi z}{\pi z} \right)^2$$

We get that  $B(z) = H(z) + K(z)$  by [8, p. 191]. Then as  $b(z) = -B(-z)$ ,  $K$  is even and  $H$  is odd, we get that

$$b(z) = -(H(-z) + K(-z)) = -H(-z) - K(z) = H(z) - K(z)$$

As a result

$$\begin{aligned} \widehat{m}_{\delta,[-L,L]}(x) &= \frac{1}{2} \left( H(\widehat{\delta(\cdot - L)})(x) + H(\widehat{\delta(L - \cdot)})(x) \right) - \frac{1}{2} \left( K(\widehat{\delta(\cdot - L)})(x) + K(\widehat{\delta(L - \cdot)})(x) \right) \\ &= \frac{1}{2} \left( \frac{e^{iLx}}{\delta} \widehat{H}\left(\frac{x}{\delta}\right) + \frac{e^{-iLx}}{\delta} \widehat{H}\left(\frac{-x}{\delta}\right) \right) - \frac{1}{2} \left( \frac{e^{iLx}}{\delta} \widehat{K}\left(\frac{x}{\delta}\right) + \frac{e^{-iLx}}{\delta} \widehat{K}\left(\frac{-x}{\delta}\right) \right) \\ &= \frac{i \sin(Lx)}{\delta} \widehat{H}\left(\frac{x}{\delta}\right) - \frac{\cos(Lx)}{\delta} \widehat{K}\left(\frac{x}{\delta}\right) \end{aligned}$$

Next, note that

$$\widehat{K}(x) = \begin{cases} \left(1 - \frac{|x|}{2\pi\delta}\right) & \text{if } |x| \leq 2\pi \\ 0 & \text{otherwise} \end{cases}$$

Moreover [8, (2.28)] gives us that if  $J(z) = \frac{1}{2}H'(z)$ ,

$$\widehat{J}(x) = \begin{cases} 1 & x = 0 \\ \frac{1}{2}x \left(1 - \frac{|x|}{2\pi}\right) \cot\left(\frac{x}{2}\right) + \frac{|x|}{2\pi} & 0 < |x| \leq 2\pi \\ 0 & |x| \geq 2\pi \end{cases}$$

Hence

$$\widehat{H}(x) = \frac{\widehat{H'}(x)}{ix} = 2 \frac{\widehat{J}(x)}{ix}$$

Putting it together, we get that

$$\widehat{m}_{\delta,[-L,L]}(x) = \begin{cases} \frac{\sin(Lx)}{\delta} \left(1 - \frac{|x|}{2\pi\delta}\right) \cot\left(\frac{x}{2\delta}\right) + \frac{\operatorname{sgn}(x) \sin(Lx)}{\pi\delta} - \frac{\cos(Lx)}{\delta} \left(1 - \frac{|x|}{2\pi\delta}\right) & \text{if } |x| \leq 2\pi\delta \\ 0 & \text{otherwise} \end{cases}$$

Then the result follows.  $\square$

To bound the integral term rigorously, let  $N$  be a large constant (e.g.  $N = 10^5$  and  $N > L$ ), then we attempt to compute the integral numerically by splitting it up like the following

$$\begin{aligned} & -\frac{1}{2\pi} \int_{-\infty}^{\infty} \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa}{2} + \frac{it}{2} \right) m_{\delta, [-L, L]}(t) dt \\ &= -\frac{1}{\pi} \int_{0 \leq t \leq N} \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa}{2} + \frac{it}{2} \right) m_{\delta, [-L, L]}(t) dt + \frac{1}{\pi} \int_{N < t} \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa}{2} + \frac{it}{2} \right) (\chi_{[-L, L]}(t) - m_{\delta, [-L, L]}(t)) dt \end{aligned}$$

We compute the first integral numerically, for the latter integral, we deal with it with the following two lemmas.

**Lemma 3.13.** For all  $x$ , we have that

$$|\chi_{[-L, L]}(x) - m_{\delta, [-L, L]}(x)| \leq \frac{1}{\pi^2 \delta^2 (L - x)^2} + \frac{1}{\pi^2 \delta^2 (L + x)^2}$$

*Proof.* Using [8, p. 2.24], we have that  $|\operatorname{sgn}(x) - H(x)| \leq K(x)$ . As a result

$$\begin{aligned} |\operatorname{sgn}(x) - b(x)| &= |\operatorname{sgn}(x) - (H(x) - K(x))| \\ &\leq 2K(x) \end{aligned}$$

Therefore

$$\begin{aligned} |\chi_{[-L, L]}(x) - m_{\delta, [-L, L]}(x)| &\leq \frac{1}{2} (|\operatorname{sgn}(x + L) - b(\delta(x + L))| + |\operatorname{sgn}(L - x) - b(\delta(L - x))|) \\ &= \frac{1}{2} (|\operatorname{sgn}(\delta(x + L)) - b(\delta(x + L))| + |\operatorname{sgn}(\delta(L - x)) - b(\delta(L - x))|) \\ &\leq K(\delta(L - x)) + K(\delta(L + x)) \\ &= \frac{1}{\pi^2 \delta^2 (L - x)^2} + \frac{1}{\pi^2 \delta^2 (L + x)^2} \end{aligned}$$

So we are done □

**Lemma 3.14.**

$$\begin{aligned} & -\frac{1}{2\pi} \int_{-\infty}^{\infty} \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa}{2} + \frac{it}{2} \right) m_{\delta, [-L, L]}(t) dt \\ &\leq -\frac{1}{\pi} \int_{0 \leq t \leq N} \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa}{2} + \frac{it}{2} \right) m_{\delta, [-L, L]}(t) dt \\ &+ \frac{1}{\pi^3 \delta^2} \left( \frac{N \log((1/2 + \kappa)^2 + N^2)}{N^2 - L^2} + \frac{1}{(1/2 + \kappa)^2 + N^2} \left( L \log\left(\frac{N + L}{N - L}\right) + 2(1/2 + \kappa) \arctan\left(\frac{1/2 + \kappa}{N}\right) \right) \right) \end{aligned}$$

*Proof.* Combine the above lemma with the fact that by lemma 2.7 of [6], we have that

$$\Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa + it}{2} \right) \leq \frac{1}{2} \log((1/2 + \kappa)^2 + t^2) - \gamma$$

we get

$$\begin{aligned} & \frac{1}{\pi} \int_{N < t} \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa}{2} + \frac{it}{2} \right) (\chi_{[-L, L]}(t) - m_{\delta, [-L, L]}(t)) dt \\ & \leq \frac{1}{\pi} \int_{N < t} \left( \frac{1}{2} \log((1/2 + \kappa)^2 + t^2) - \gamma \right) \left( \frac{1}{\pi^2 \delta^2 (L - x)^2} + \frac{1}{\pi^2 \delta^2 (x + L)^2} \right) dt \end{aligned}$$

Note, clearly

$$\begin{aligned} -\gamma \int_{N < t} \frac{1}{(L - t)^2} + \frac{1}{(t + L)^2} &= -\gamma \left( \frac{1}{L - t} - \frac{1}{L + t} \Big|_N^\infty \right) \\ &= -\gamma \left( +\frac{1}{N - L} + \frac{1}{L + N} \right) \\ &= \frac{-2\gamma N}{N^2 - L^2} \end{aligned}$$

On the other hand, integration by parts gets us that

$$\begin{aligned} & \int_N^\infty \frac{1}{2} \log((1/2 + \kappa)^2 + t^2) \left( \frac{1}{(L - t)^2} + \frac{1}{(L + t)^2} \right) dt \\ &= \frac{1}{2} \log((1/2 + \kappa)^2 + t^2) \frac{-2t}{t^2 - L^2} \Big|_N^\infty + \int_N^\infty \frac{2t^2}{(t^2 - L^2)(t^2 + (1/2 + \kappa)^2)} dt \\ &= \frac{N \log((1/2 + \kappa)^2 + N^2)}{N^2 - L^2} + \frac{1}{(1/2 + \kappa)^2 + N^2} \int_N^\infty \left( \frac{2L^2}{t^2 - L^2} + \frac{2(1/2 + \kappa)^2}{t^2 + (1/2 + \kappa)^2} \right) dt \\ &= \frac{N \log((1/2 + \kappa)^2 + N^2)}{N^2 - L^2} + \frac{1}{(1/2 + \kappa)^2 + N^2} \left( L \log\left(\frac{N + L}{N - L}\right) + 2(1/2 + \kappa) \arctan\left(\frac{1/2 + \kappa}{N}\right) \right) \end{aligned}$$

From here, the result follows.  $\square$

*proof of Theorem 1.2.* Using the lemma 3.14 from above and eq. (2), we get, and assuming there are no zeros of height less than  $L$ , we get that

$$\begin{aligned} & \frac{\widehat{m}_{\delta, [-L, L]}(0)}{2\pi} \log(q) \\ & \leq \frac{\widehat{m}_{\delta, [-L, L]}(0)}{2\pi} \log(\pi) \\ & + \frac{1}{\pi} \sum_{n \leq e^{2\pi\delta}} \frac{\Lambda(n) |\widehat{m}_{\delta, [-L, L]}(\log(n))|}{\sqrt{n}} \\ & - \frac{1}{\pi} \int_{0 \leq t \leq N} \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa}{2} + \frac{it}{2} \right) m_{\delta, [-L, L]}(t) dt \\ & + \frac{1}{\pi^3 \delta^2} \left( \frac{N \log((1/2 + \kappa)^2 + N^2)}{N^2 - L^2} + \frac{1}{(1/2 + \kappa)^2 + N^2} \left( L \log\left(\frac{N + L}{N - L}\right) + 2(1/2 + \kappa) \arctan\left(\frac{1/2 + \kappa}{N}\right) \right) \right) \end{aligned}$$

Ensuring  $\widehat{m}_{\delta, [-L, L]}(0) > 0$ , let  $\log(q_0)$  be the RHS of the above equation after dividing by  $\frac{\widehat{m}_{\delta, [-L, L]}(0)}{2\pi}$ , then note

- For  $L = 4$ ,  $\delta = 0.2$ , we get  $\log(q_0) = 4.1150060 + 3.36 \cdot 10^{-8}$ .
- For  $L = 3$ ,  $\delta = 0.3$ , we get  $\log(q_0) = 5.0758697 + 1.65 \cdot 10^{-8}$ .
- For  $L = 2$ ,  $\delta = 0.5$ , we get  $\log(q_0) = 7.8241763 + 2.65 \cdot 10^{-8}$ .
- For  $L = 1$ ,  $\delta = 0.9$ , we get  $\log(q_0) = 23.069395 + 2.27 \cdot 10^{-7}$

The theorem then follows by exponentiating. □

Then Corollary 1.3 follows by using a simple GP/Pari script that computes the lowest zeros of  $L(s, \chi)$  for each conductor  $q$  up to  $q_0$ .

## 3.6 Proof of Theorem 1.4

### 3.6.1 Study Prime Sum

**Lemma 3.15.** For all  $x \in \mathbb{R}$

$$|\widehat{m}_{\delta, [\alpha, \beta]}(x)| \leq \beta - \alpha + \frac{1}{\delta}$$

*Proof.* Note that using proposition 3.11, we have

$$\begin{aligned} |\widehat{m}_{\delta, [\alpha, \beta]}(x)| &\leq |\widehat{m}_{\delta, [\alpha, \beta]}(x) - \widehat{\chi_{[\alpha, \beta]}}(x)| + |\widehat{\chi_{[\alpha, \beta]}}(x)| \\ &\leq \int_{-\infty}^{\infty} |(m_{\delta, [\alpha, \beta]}(t) - \chi_{[\alpha, \beta]}(t))e^{-itx}| dt + \int_{\alpha}^{\beta} 1 dt \\ &= \frac{1}{\delta} + \beta - \alpha \end{aligned}$$

□

**Lemma 3.16.** Let  $\gamma = \frac{\alpha + \beta}{2}$  and  $\Delta = \frac{\beta - \alpha}{2}$ , then

$$\widehat{m}_{\delta, [\alpha, \beta]}(x) = \frac{e^{-i\gamma x}}{\delta} \left( 2 \sin(\Delta x) \frac{\widehat{J}\left(\frac{x}{\delta}\right)}{\frac{x}{\delta}} - \cos(\Delta x) \widehat{K}\left(\frac{x}{\delta}\right) \right)$$

*Proof.*

$$\begin{aligned}
& \widehat{m}_{\delta, [\alpha, \beta]}(x) \\
&= \frac{1}{2} \left( H(\widehat{\delta(\beta - \cdot)})(x) + H(\widehat{\delta(-\alpha)})(x) \right) - \frac{1}{2} \left( K(\widehat{\delta(\beta - \cdot)})(x) + K(\widehat{\delta(-\alpha)})(x) \right) \\
&= \frac{1}{2} \left( H(\widehat{\delta(\Delta + \gamma - \cdot)})(x) + H(\widehat{\delta(-(\gamma - \Delta))})(x) \right) - \frac{1}{2} \left( K(\widehat{\delta(\gamma + \Delta - \cdot)})(x) + K(\widehat{\delta(-(\gamma - \Delta))})(x) \right) \\
&= \frac{1}{2} \left( \frac{e^{-i(\Delta + \gamma)x} \widehat{H}\left(\frac{-x}{\delta}\right)}{\delta} + \frac{e^{i(\gamma - \Delta)x} \widehat{H}\left(\frac{x}{\delta}\right)}{\delta} \right) - \frac{1}{2} \left( \frac{e^{-i(\gamma + \Delta)x} \widehat{K}\left(\frac{-x}{\delta}\right)}{\delta} + \frac{e^{i(\gamma - \Delta)x} \widehat{K}\left(\frac{x}{\delta}\right)}{\delta} \right) \\
&= \frac{e^{-i\gamma x}}{\delta} \left( i \sin(\Delta x) \widehat{H}\left(\frac{x}{\delta}\right) - \cos(\Delta x) \widehat{K}\left(\frac{x}{\delta}\right) \right) \\
&= \frac{e^{-i\gamma x}}{\delta} \left( 2 \sin(\Delta x) \widehat{J}\left(\frac{x}{\delta}\right) - \cos(\Delta x) \widehat{K}\left(\frac{x}{\delta}\right) \right)
\end{aligned}$$

□

**Lemma 3.17.** Assuming RH, for all  $x > 73.2$

$$\begin{aligned}
& \sum_{n \leq x} \frac{\Lambda(n)}{\sqrt{n}} \\
&= \sum_{n \leq 73.2} \frac{\Lambda(n)}{\sqrt{n}} + 2\sqrt{x} + \frac{\log^2(x)}{8\pi} + \frac{1}{48\pi} (\log^3(x) - \log^3(73.2)) - \sqrt{73.2} - \frac{\psi(73.2)}{\sqrt{73.2}}
\end{aligned}$$

*Proof.* Under RH, by [4, (1)], for all  $x \geq 73.2$

$$|\psi(x) - x| \leq \left( \frac{\log(x)}{8\pi} \right) \sqrt{x} \log(x)$$

As a result, we have that

$$\begin{aligned}
\sum_{n \leq x} \frac{\Lambda(n)}{\sqrt{n}} &= \sum_{n \leq 73.2} \frac{\Lambda(n)}{\sqrt{n}} + \sum_{73.2 < n \leq x} \frac{\Lambda(n)}{\sqrt{n}} \\
&= \sum_{n \leq 73.2} \frac{\Lambda(n)}{\sqrt{n}} + \int_{73.2}^x \frac{1}{\sqrt{t}} d\psi \\
&= \sum_{n \leq 73.2} \frac{\Lambda(n)}{\sqrt{n}} + \frac{\psi(x)}{\sqrt{x}} - \frac{\psi(73.2)}{\sqrt{73.2}} + \frac{1}{2} \int_{73.2}^x \frac{\psi(t)}{t^{3/2}} dt \\
&= \sum_{n \leq 73.2} \frac{\Lambda(n)}{\sqrt{n}} + \frac{\psi(x) - x}{\sqrt{x}} + \sqrt{x} - \frac{\psi(73.2)}{\sqrt{73.2}} + \frac{1}{2} \int_{73.2}^x \frac{\psi(t) - t}{t^{3/2}} dt + \frac{1}{2} \int_{73.2}^x \frac{1}{\sqrt{t}} dt \\
&\leq \sum_{n \leq 73.2} \frac{\Lambda(n)}{\sqrt{n}} + \sqrt{x} + \frac{\log^2(x)}{8\pi} - \frac{\psi(73.2)}{\sqrt{73.2}} + \frac{1}{2} \int_{73.2}^x \frac{1}{8\pi} \frac{\log^2(t)}{t} dt + \sqrt{x} - \sqrt{73.2} \\
&= \sum_{n \leq 73.2} \frac{\Lambda(n)}{\sqrt{n}} + 2\sqrt{x} + \frac{\log^2(x)}{8\pi} + \frac{1}{48\pi} (\log^3(x) - \log^3(73.2)) - \sqrt{73.2} - \frac{\psi(73.2)}{\sqrt{73.2}}
\end{aligned}$$

□

### 3.6.2 Study Integral Term

Observe that

$$\begin{aligned}
& -\frac{1}{2\pi} \int_{-\infty}^{\infty} \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa}{2} + \frac{it}{2} \right) m_{\delta, [\alpha, \beta]}(t) dt \\
&= -\frac{1}{2\pi} \int_0^{\infty} \frac{e^{-u}}{u} \widehat{m}_{\delta, [\alpha, \beta]}(0) - \Re \widehat{m}_{\delta, [\alpha, \beta]} \left( \frac{u}{2} \right) \frac{e^{-(\frac{1/2 + \kappa}{2})u}}{1 - e^{-u}} du \\
&= -\frac{1}{2\pi} \int_0^{4\pi\delta} \frac{e^{-u}}{u} \widehat{m}_{\delta, [\alpha, \beta]}(0) - \Re \widehat{m}_{\delta, [\alpha, \beta]} \left( \frac{u}{2} \right) \frac{e^{-(\frac{1/2 + \kappa}{2})u}}{1 - e^{-u}} du - \frac{\widehat{m}_{\delta, [\alpha, \beta]}(0)}{2\pi\delta} E_1(4\pi\delta) \\
&= -\frac{\widehat{m}_{\delta, [\alpha, \beta]}(0)}{2\pi} \int_0^{4\pi\delta} \frac{e^{-u}}{u} - \frac{e^{-(\frac{1/2 + \kappa}{2})u}}{1 - e^{-u}} du \\
&+ \frac{1}{2\pi} \int_0^{4\pi\delta} \left( \Re \widehat{m}_{\delta, [\alpha, \beta]}(u/2) - \widehat{m}_{\delta, [\alpha, \beta]}(0) \right) \frac{e^{-(\frac{1/2 + \kappa}{2})u}}{1 - e^{-u}} du - \frac{\widehat{m}_{\delta, [\alpha, \beta]}(0)}{2\pi\delta} E_1(4\pi\delta)
\end{aligned}$$

We now have to upper bound the second integral though the following lemmas.

**Lemma 3.18.** For all  $x \in (0, 4\pi\delta)$

$$\begin{aligned}
\Re \widehat{m}_{\delta, [\alpha, \beta]} - \Re \widehat{m}_{\delta, [\alpha, \beta]}(0) &\leq \frac{\alpha^2 x^2}{4\delta} + \frac{\beta^2 x^2}{4\delta} + \frac{x}{2\pi\delta^2} \\
&+ \frac{(\beta - \alpha)x}{2\pi\delta}
\end{aligned}$$

*Proof.* Let  $\Delta = \frac{\beta - \alpha}{2}$  and  $\gamma = \frac{\beta + \alpha}{2}$ . Recall that  $\widehat{m}_{\delta, [\alpha, \beta]}(x) = \frac{e^{-i\gamma x}}{\delta} \left( 2 \sin(\Delta x) \widehat{J}\left(\frac{x}{\delta}\right) - \cos(\Delta x) \widehat{K}\left(\frac{x}{\delta}\right) \right)$  and  $\widehat{m}(0) = 2\Delta - \frac{1}{\delta}$ . As a result

$$\begin{aligned}
\Re \widehat{m}_{\delta, [\alpha, \beta]}(x) - \widehat{m}_{\delta, [\alpha, \beta]}(0) &= \frac{1 - \cos(\gamma x) \cos(\Delta x) \widehat{K}\left(\frac{x}{\delta}\right)}{\delta} \\
&+ 2 \left( \cos(\gamma x) \frac{\sin \Delta x}{x} \widehat{J}\left(\frac{x}{\delta}\right) - \Delta \right)
\end{aligned}$$

For the first term, note that  $\cos(\gamma x) \cos(\Delta x) = \frac{1}{2}(\cos(\alpha x) + \cos(\beta x))$  and  $\cos(x) \geq 1 - \frac{x^2}{2}$ , so then the first line is bounded above by

$$\frac{(\alpha^2 + \beta^2)x^2}{4\delta} + \frac{x}{2\pi\delta^2}$$

For the second line, we can re-write it as

$$\begin{aligned}
& \cos(\delta x) \frac{\sin(\Delta x)}{x} \left( \frac{1}{2} \frac{x}{\delta} \cot\left(\frac{x}{2\delta}\right) \left(1 - \frac{x}{2\pi\delta}\right) + \frac{x}{2\pi\delta} \right) - \Delta \\
&= \frac{\cos(\gamma x) \sin(\Delta x)}{2\pi\delta} + \Delta \left( \cos(\gamma x) \operatorname{sinc}(\Delta x) \frac{\cos(\frac{x}{2\delta})}{\operatorname{sinc}(\frac{x}{2\delta})} - 1 \right) - \frac{\Delta x}{2\pi\delta} \cos(\gamma x) \operatorname{sinc}(\Delta x) \frac{\cos(\frac{x}{2\delta})}{\operatorname{sinc}(\frac{x}{2\delta})} \\
&= \frac{\sin(\beta x) - \sin(\alpha x)}{4\pi\delta} + \Delta \left( \cos(\gamma x) \operatorname{sinc}(\Delta x) \frac{\cos(\frac{x}{2\delta})}{\operatorname{sinc}(\frac{x}{2\delta})} - 1 \right) - \frac{\Delta x}{2\pi\delta} \cos(\gamma x) \operatorname{sinc}(\Delta x) \frac{\cos(\frac{x}{2\delta})}{\operatorname{sinc}(\frac{x}{2\delta})} \\
&\leq \frac{\beta - \alpha}{4\pi\delta} + \frac{(\beta - \alpha)x}{4\pi\delta} \\
&= \frac{(\beta - \alpha)x}{2\pi\delta}
\end{aligned}$$

□

**Lemma 3.19.** For all  $x > 0$  and  $a > 0$

$$\int_0^x \frac{te^{-at}}{1 - e^{-t}} dt \leq x + \frac{x^2}{2}$$

$$\int_0^x \frac{t^2 e^{-at}}{1 - e^{-t}} dt \leq \frac{x^2}{2} + \frac{x^3}{3}$$

*Proof.* In both integrals, we trivially dropped the  $e^{-at}$  by 1. Then note for  $t > 0$ ,  $e^t \geq 1 + t$  implies  $e^{-t} \leq \frac{1}{1+t}$ , so  $1 - e^{-t} \geq \frac{t}{t+1}$ . Therefore

$$\begin{aligned}
\frac{t}{1 - e^{-t}} &\leq 1 + t \\
\frac{t^2}{1 - e^{-t}} &\leq t + t^2
\end{aligned}$$

Then the result follows trivially by integration. □

**Proposition 3.20.**

$$\begin{aligned}
& \frac{1}{2\pi} \int_{-\infty}^{\infty} \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa}{2} + \frac{it}{2} \right) m_{\delta, [\alpha, \beta]}(t) dt \\
&\leq -\frac{\widehat{m}_{\delta, [\alpha, \beta]}(0)}{2\pi} \int_0^{4\pi\delta} \frac{e^{-u}}{u} - \frac{e^{-(\frac{1/2+\kappa}{2})u}}{1 - e^{-u}} du \\
&+ \frac{1}{2\pi} \left( \frac{(\alpha^2 + \beta^2)}{16\delta} \left( \frac{(4\pi\delta)^2}{2} + \frac{(4\pi\delta)^3}{3} \right) + \left( \frac{1}{4\pi\delta^2} + \frac{\beta - \alpha}{4\pi\delta} \right) \left( 4\pi\delta + \frac{(4\pi\delta)^2}{2} \right) \right) \\
&- \frac{\widehat{m}_{\delta, [\alpha, \beta]}(0)}{2\pi\delta} E_1(4\pi\delta)
\end{aligned}$$

*Proof.* All we needed to do is to bound

$$\frac{1}{2\pi} \int_0^{4\pi\delta} (\Re \hat{m}_{\delta, [\alpha, \beta]}(u/2) - \hat{m}_{\delta, [\alpha, \beta]}(0)) \frac{e^{-(\frac{1/2+\kappa}{2})u}}{1 - e^{-u}} du$$

We get that by lemma 3.18

$$\Re \hat{m}_{\delta, [\alpha, \beta]}(u/2) - \hat{m}_{\delta, [\alpha, \beta]}(0) \leq \frac{(\alpha^2 + \beta^2)u^2}{16\delta} + \frac{u}{4\pi\delta^2} + \frac{(\beta - \alpha)u}{4\pi\delta}$$

From here, using lemma 3.19 gets us that integral is bounded above by

$$\frac{1}{2\pi} \left( \frac{(\alpha^2 + \beta^2)}{16\delta} \left( \frac{(4\pi\delta)^2}{2} + \frac{(4\pi\delta)^3}{3} \right) + \left( \frac{1}{4\pi\delta^2} + \frac{\beta - \alpha}{4\pi\delta} \right) (4\pi\delta + \frac{(4\pi\delta)^2}{2}) \right)$$

□

*proof of Theorem 1.4.* Let  $\alpha = \gamma_{\chi, n}$  and  $\beta = \gamma_{\chi, n+1}$ . Let  $\epsilon > 0$  and  $\delta = (1 + \epsilon) \frac{1}{\beta - \alpha}$ . Using non-negativity and the fact there are no-zeros of  $L$  that is inbetween  $\gamma_{\chi, n} < \gamma_{\chi, n+1}$ :

$$\begin{aligned} \frac{\hat{m}_{\delta, [\alpha, \beta]}(0)}{2\pi} \log \left( \frac{q}{\pi} \right) &\leq + \frac{1}{2\pi} \sum_{n \in \mathbb{N}} \Lambda(n) \left[ \frac{\chi(n) \hat{m}_{\delta, [-L, L]}(\log(n))}{\sqrt{n}} + \frac{\bar{\chi}(n) \hat{m}_{\delta, [-L, L]}(-\log(n))}{\sqrt{n}} \right] \\ &\quad - \frac{1}{2\pi} \int_{-\infty}^{\infty} \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa}{2} + \frac{it}{2} \right) m_{\delta, [-L, L]}(t) dt \end{aligned}$$

This gets us

$$\begin{aligned} \frac{\hat{m}_{\delta, [\alpha, \beta]}(0)}{2\pi} \log \left( \frac{q}{\pi} \right) &\leq \frac{1}{\pi} \sum_{n \leq e^{2\pi\delta}} \frac{\Lambda(n)}{\sqrt{n}} |\hat{m}_{\delta, [\alpha, \beta]}(\log(n))| \\ &\quad - \frac{1}{2\pi} \int_{-\infty}^{\infty} \Re \frac{\Gamma'}{\Gamma} \left( \frac{1/2 + \kappa}{2} + \frac{it}{2} \right) m_{\delta, [-L, L]}(t) dt \end{aligned}$$

Now, using the upper bounds on the prime sum and integral that we obtain earlier, we get that (the  $o(1)$  term is going to zero as

$$\begin{aligned} \frac{\beta - \alpha - \frac{1}{\delta}}{2\pi} \log \left( \frac{q}{\pi} \right) &\leq \frac{\beta - \alpha + \frac{1}{\delta}}{\pi} (2 + o_{\delta \rightarrow \infty}(1) \cap O_{\mathbb{R}_{>0}}(1)) e^{\pi\delta} \\ &\leq - \frac{\beta - \alpha - \frac{1}{\delta}}{2\pi} \int_0^{4\pi\delta} \frac{e^{-u}}{u} - \frac{e^{-(\frac{1/2+\kappa}{2})u}}{1 - e^{-u}} du \\ &\quad + \frac{1}{2\pi} \left( \frac{(\alpha^2 + \beta^2)}{16\delta} \left( \frac{(4\pi\delta)^2}{2} + \frac{(4\pi\delta)^3}{3} \right) + \left( \frac{1}{4\pi\delta^2} + \frac{\beta - \alpha}{4\pi\delta} \right) (4\pi\delta + \frac{(4\pi\delta)^2}{2}) \right) \end{aligned}$$

Now dividing through by  $\frac{\beta-\alpha-\frac{1}{\delta}}{2\pi}$  and using the fact that  $\delta(\beta-\alpha) = 1+\epsilon$  and  $\frac{\beta-\alpha+1/\delta}{\beta-\alpha-1/\delta} = 1 + \frac{2}{\epsilon}$ , we get

$$\begin{aligned} \log\left(\frac{q}{\pi}\right) &\leq \left(1 + \frac{2}{\epsilon}\right)(4 + o_{\delta \rightarrow \infty, \epsilon}(1) \cap O_{\mathbb{R}_{>0}, \epsilon}(1))e^{\pi\delta} \\ &\quad + \frac{1}{\epsilon} \left( \frac{(\alpha^2 + \beta^2)}{16} \left( \frac{(4\pi\delta)^2}{2} + \frac{(4\pi\delta)^3}{3} \right) + \left( \frac{1}{4\pi} + \frac{1+\epsilon}{4\pi} \right) \left( 4\pi + \frac{(4\pi)^2\delta}{2} \right) \right) \end{aligned}$$

Simplifying a bit more

$$\begin{aligned} \log\left(\frac{q}{\pi}\right) &\leq \left(1 + \frac{2}{\epsilon}\right)(4 + o_{\delta \rightarrow \infty, \epsilon}(1) \cap O_{\mathbb{R}_{>0}, \epsilon}(1))e^{\pi\delta} \\ &\quad + \frac{1}{\epsilon} \frac{(\alpha^2 + \beta^2)}{16} \left( \frac{(4\pi\delta)^2}{2} + \frac{(4\pi\delta)^3}{3} \right) \end{aligned}$$

Then a bit more gives

$$\log\left(\frac{q}{\pi}\right) \leq \left(1 + \frac{2}{\epsilon}\right)(4 + o_{\delta \rightarrow \infty, \epsilon}(1) \cap O_{\mathbb{R}_{>0}, \epsilon}(1) + \frac{\alpha^2 + \beta^2}{16(2+\epsilon)} \left( \frac{(2\pi\delta)^2}{2} + \frac{(2\pi\delta)^3}{3} \right) e^{-\pi\delta}) e^{\pi\delta}$$

Taking out  $-\log(\pi)$  to combine with the big O terms to the other side gives us, and upper bounding the coefficient on  $\alpha^2 + \beta^2$  that is not divided by 16 by 33 gets us

$$\log(q) \leq \left(4 + \frac{8}{\epsilon} + o_{\delta \rightarrow \infty, \epsilon}(1) \cap O_{\mathbb{R}_{>0}, \epsilon}(1) + \frac{33(\alpha^2 + \beta^2)}{16} (1 + 2/\epsilon) \frac{1}{2+\epsilon}\right) e^{\pi\delta}$$

Taking logarithm gets us

$$\log \log(q) \leq \log\left(4 + \frac{8}{\epsilon} + (1 + 2/\epsilon) o_{\delta \rightarrow \infty}(1) \cap O_{\delta \in \mathbb{R}_{>0}}(1) + \frac{33(\alpha^2 + \beta^2)}{16} (1 + 2/\epsilon) \frac{1}{2+\epsilon}\right) + \pi\delta$$

Now, for a fixed  $n$ , observe that by the Riemann von mongolt formula, as  $q \rightarrow \infty$ ,  $|\gamma_{\chi, n}|, |\gamma_{\chi, n+1}| \leq 1$ . This means that for  $n$  fixed,  $\alpha^2 + \beta^2$  is bounded. Therefore then when  $q$  is sufficiently large, we get

$$\log \log(q) \leq \log\left(4 + \frac{8}{\epsilon} + (1 + 2/\epsilon) o_{\delta \rightarrow \infty}(1) \cap O_{\delta \in \mathbb{R}_{>0}}(1) + O_{q \in \mathbb{N}}(1) (1 + 2/\epsilon) \frac{1}{2+\epsilon}\right) + \pi(1 + \epsilon) \frac{1}{\gamma_{\chi, n+1} - \chi_n}$$

So for  $q$  large enough, we can rearrange and invert to get

$$\gamma_{\chi, n+1} - \gamma_{\chi, n} \leq \pi(1 + \epsilon) \frac{1}{\log \log q - \log\left(4 + \frac{8}{\epsilon} + (1 + 2/\epsilon) O_n(1)\right)}$$

In fact, if we let  $\epsilon = \frac{1}{\log \log q}$ , we should get for  $q$  sufficiently large

$$\begin{aligned} & \log \log(q) \\ & \leq \log(4 + 8 \log \log(q) + (1 + 2 \log \log(q))o_{\delta \rightarrow \infty}(1) \cap O_{\delta \in \mathbb{R}_{>0}}(1) + O_{q \in \mathbb{N}}(1)(1 + 2 \log \log q) \frac{1}{2 + \frac{1}{\log \log q}}) \\ & + \pi(1 + \frac{1}{\log \log q}) \frac{1}{\gamma_{\chi, n+1} - \chi_n} \end{aligned}$$

In fact, if we let  $\epsilon = \frac{1}{\log \log q}$ , we should get for  $q$  sufficiently large

$$\begin{aligned} & \log \log(q) \\ & \leq \log(4 + 8 \log \log(q) + (1 + 2 \log \log(q))o_{\delta \rightarrow \infty}(1) \cap O_{\delta \in \mathbb{R}_{>0}}(1) + O_{q \in \mathbb{N}}(1)(1 + 2 \log \log q) \frac{1}{2 + \frac{1}{\log \log q}}) \\ & + \pi(1 + \frac{1}{\log \log q}) \frac{1}{\gamma_{\chi, n+1} - \chi_n} \end{aligned}$$

Therefore, for  $q$  sufficiently large, we can invert to get

$$\gamma_{\chi, n+1} - \gamma_{\chi, n} \leq \pi(1 + \frac{1}{\log \log q}) \frac{1}{\log \log q - \log(4 + 8 \log \log q + (1 + 2 \log \log q)O_n(1))}$$

This means that

$$\begin{aligned} \gamma_{\chi, n+1} - \gamma_{\chi, n} & \leq \pi(1 + \frac{1}{\log \log q} + o(1)) \frac{1}{\log \log q} \quad q \rightarrow \infty \\ & \leq (\pi + o(1)) \frac{1}{\log \log q} \quad q \rightarrow \infty \end{aligned}$$

□

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